

Nonlinear ill-posed problem analysis in model-based parameter estimation and experimental design



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ABSTRACT

Discrete ill-posed problems are often encountered in engineering applications. Still, their sound analysis is not yet common practice and difficulties arising in the determination of uncertain parameters are typically not assigned properly. This contribution provides a tutorial review on methods for identifiability analysis, regularization techniques and optimal experimental design. A guideline for the analysis and classification of nonlinear ill-posed problems to detect practical identifiability problems is given. Techniques for the regularization of experimental design problems resulting from ill-posed parameter estimations are discussed. Applications are presented for three different case studies of increasing complexity.

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1. Introduction

Typical causes of poor or even no identifiability in nonlinear (and linear) system identification are: over-parameterized process models, model structures not matching the system, limited experimental information (either in quantity or quality), correlations in parameters or experimental data, parameters with little or null influence on observed variables and inappropriate initial parameter guesses (Bard, 1974; Bates and Watts, 1988; Franceschini and Macchietto, 2008; Vajda et al., 1989; Walter and Pronzato, 1996). This generates problems in parameter estimation (PE), e.g., multiple, meaningless, inaccurate or unstable solutions, and/or convergence and numerical problems in the solver. Those problems lead for instance, to a model with poor extrapolation properties or parameters with large variance or uncertainty (the so called non-identifiable parameters).

According to Hadamard (1923), a well-posed problem means existence and uniqueness of its solution, as well as continuous dependence of the solution on the data. In other words, if the solution does not exist, or if it is not unique, or if a small change of the

experimental data produces a large perturbation of the solution, the problem is ill-posed (Bertero et al., 1988; Hansen, 1998, 2007; Hoerl and Kennard, 1970a; Marquardt, 1970). Continuity is related to the requirement of stability or robustness of the solution (Bertero et al., 1988). However, this condition is necessary for stability, but it is not sufficient as especially for discrete ill-posed problems (coming from discrete available data or discretization of ill-posed problems) lack of robustness against noise is usually evidenced (Bertero et al., 1988). Accordingly in PE, besides the strong relationship between identifiability and ill-posedness of a problem, the error propagation from data to the solution should also be addressed.

An analysis of the ill-posedness of a PE problem could be related to the ill-conditioning and thus, to the numerical stability and linear dependencies of a specific matrix, in the sense that noisy measurement data may lead to significant misinterpretations of the solution (Kaltenbacher et al., 2008). In linear estimation (e.g. the linear regression model $y = X\beta + \varepsilon$) the problem could be considered ill-conditioned and then ill-posed if the data matrix X is ill-conditioned (Belsley et al., 1980; Hansen, 1998, 2007; Hoerl and Kennard, 1970a; Marquardt, 1970). This ill-conditioning (also called collinearity) is of paramount importance to the efficacy of least-squares estimation though it is a non-statistical problem (Belsley et al., 1980). For instance, in the presence of collinearity the uniqueness of the least-squares estimator may not be guaranteed.

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List of abbreviations

DAEs	differential and algebraic equations
FIM	Fisher information matrix
IG	initial guess
None	problem without regularization
ODEs	ordinary differential equations
OED	optimal experimental design
PE	parameter estimation
QRP	QR decomposition with column pivoting
Reg	regularization technique either SsS, TSVD or Tikh
SsS	identifiable parameter subset selection based-regularization
SVD	singular value decomposition
SVs	singular value spectrum
Tikh	Tikhonov based-regularization
TSVD	truncated singular value decomposition based-regularization

List of symbols

C	parameter variance-covariance matrix without regularization
C^{Reg}	Parameter variance-covariance matrix with regularization <i>Reg</i>
C_M	measurement error variance-covariance matrix
f	set of DAE representing the process model
h	set of relations between the measured variables and the dependent state variables
H_θ	simplified Hessian matrix (FIM)
H_θ^{Reg}	simplified Hessian matrix (FIM) after applying the regularization <i>Reg</i>
J_θ	Jacobian matrix
J_θ^{Reg}	Jacobian matrix after applying the regularization <i>Reg</i>
L	operator matrix in Tikhonov regularization, i.e., <i>Reg</i> = <i>Tikh</i>
N_m	number of experimental data sampling times
N_{mu}	number of input variable switching times
N_{mod}	number of available models
N_p	number of parameters
N_y	number of measured response variables
N_x	number of dependent state variables
OF	objective function of PE
P	permutation matrix of QRP decomposition
$PD(N_p)$	subspace of the positive definite matrices with dimension $N_p \times N_p$
$PSD(N_p)$	subspace of positive semi-definite matrices with dimension $N_p \times N_p$
Q	orthogonal matrix of QRP
r_ϵ	numerical ϵ -rank
R	upper triangular matrix with decreasing diagonal elements of QRP
S	sensitivity matrix
$\tilde{S}$	scaled sensitivity matrix
$\tilde{S}^{Reg}$	scaled sensitivity matrix after applying regularization <i>Reg</i>
S_v	rectangular diagonal matrix with singular values on the diagonal of SVD
$Sym(N_p)$	space of symmetric matrices with dimension $N_p \times N_p$
t	independent variable time
t_k	discrete time instance k th
u	input action or experimental design vector
u_{IG}	initial guess of experimental design vector

$\hat{u}_{crit}$	optimal experimental design vector after computing OED with $crit = \{A, D, E\}$
U	real orthogonal matrix of SVD
$var(\hat{\theta}_k)$	parameter variance of parameter estimate $\hat{\theta}_k$
V	real orthogonal matrix of SVD
x	dependent state variable
x_0	initial conditions of the dependent state variable
y	predicted response variable
y^m	measured response variable
$\bar{y}$	normalized predicted response variable
Y^m	total experimental data vector
Y	total predicted response vector
Z	PE residual vector
α	scaling factor
$\gamma(A)$	collinearity index of matrix $A = \{\tilde{S}, \tilde{S}^{Reg}\}$
γ^{max}	maximum collinearity index
ε	measurement error
ϵ	singular value threshold
ϵ_κ	singular value threshold w.r.t. maximum condition number
ϵ_γ	singular value threshold w.r.t. maximum collinearity index
$\kappa(A)$	condition number of matrix $A = \{\tilde{S}, \tilde{S}^{Reg}\}$
$\kappa^{max}(A)$	maximum condition number of matrix $A = \{S, \tilde{S}\}$
θ	parameter vector
θ_{IG}	initial guess of parameter vector
$\hat{\theta}$	unbiased parameter estimate
$\hat{\theta}^{(Np-r_\epsilon)}$	unidentifiable parameter vector after applying regularization <i>Reg</i> = <i>SsS</i>
$\tilde{\theta}^{(r_\epsilon)}$	identifiable parameter vector after applying regularization <i>Reg</i> = <i>SsS</i>
$\hat{\theta}^{(Reg)}$	regularized parameter estimate after applying regularization <i>Reg</i>
$\bar{\theta}$	normalized parameter vector
$\lambda(A)$	eigenvalue of matrix $A = \{C, H_\theta\}$
λ	Tikhonov regularization parameter
π_i	parameter variance-decomposition proportion associated with singular value ς_i
ρ	step size
σ	standard deviation of measurements
σ_i	standard deviation of measurement i
σ_i^2	variance of measurement i
ς_i	singular value i th
Σ_y	measurement error standard deviation matrix
v	step direction
Φ^{LSQ}	weighted nonlinear least squares criterion
Φ^{Reg}	weighted nonlinear least squares criterion after applying regularization <i>Reg</i>
$\mathcal{T}$	data sampling time grid
$\mathcal{T}^u$	input action time grid
Ψ	information function in OED
Ψ^{crit}	information function in OED (alphabetic criteria in OED with $crit = \{A, D, E\}$)
$\Omega(\theta)$	discrete smoothing norm functional in <i>Reg</i> = <i>Tikh</i>

In least-squares estimation for nonlinear systems ill-conditioning is locally analyzed based on the sensitivity matrix (S) of the predicted output trajectories with respect to the parameters (Asprey and Macchietto, 2000; Bard, 1974; Bates and Watts, 1988; Franceschini and Macchietto, 2008; Jacobson, 1985; Johansen, 1997; Marquardt, 1970; Vajda et al., 1989). If the sensitivity matrix is ill-conditioned, the PE problem is locally ill-posed, otherwise it is locally well-posed. For ill-conditioned problems a distinction is

made, being either rank-deficient or having an ill-determined rank (Hansen, 1998, 2007). If the numerical rank of S can be reliably calculated, the problem is considered rank-deficient, otherwise it is of ill-determined rank.

Handling ill-posed PE problems is aimed at “trying to control the parameter variance” either imposing upper and lower limits on the admissible parameter values (Carrera and Neuman, 1986) or using formal regularization techniques (Bertero et al., 1988; Kaltenbacher et al., 2008; Hansen, 1998, 2007). Although, the former seems to control the instability (at amplitude as large as that allowed by the imposed limits) the solution may remain meaningless merely oscillating between the limits (Carrera and Neuman, 1986). The latter eliminates unwarranted oscillations (with a meaningful variance level reduction) but the estimated parameters can be biased (Bardow, 2008; Haber et al., 2008, 2010; Hansen, 1998, 2007; Johansen, 1997; Marquardt, 1970).

Regularization techniques for rank-deficient problems extract the linear independent information and solve a new problem with a well-conditioned matrix. Well-known techniques are parameter subset selection (Brun et al., 2001; Burth et al., 1999; Chandrakant and Bisaria, 1998; Chu and Hahn, 2007, 2012; Golub and Van Loan, 1996; Hansen, 1998; Jörke et al., 2015; McLean et al., 2012; López et al., 2013; Vélez-Reyes, 1992; Vélez-Reyes and Verghese, 1995; Wu et al., 2011; Yao et al., 2003) and truncated singular value decomposition (TSVD) (Hansen, 1998, 2007; Xu, 1998). For ill-determined rank problems the aim is to balance the residual norm and the size of the regularized solution as there is no notion of the rank. The most common and probably the most well-known method (especially in ill-posed and inverse problems) is Tikhonov regularization (Tikhonov and Arsenin, 1977). Here a stabilizer is introduced with the idea to attract the parameters toward reasonable regions of the parameter space (Hansen, 1998, 2007; Johansen, 1997; Tikhonov and Arsenin, 1977).

Even for the best strategy, a regularized solution (estimated parameters) can remain meaningless or with low precision either because of poor information content in the experimental data, or limited data, or correlations between parameters. To further improve the precision of parameter estimates, model-based experimental design techniques have been developed with the aim to maximize the information content in experimental data (Franceschini and Macchietto, 2008; Pukelsheim, 1993). Theory and applications for the model-based optimal experimental design (OED) for well-posed problems have been addressed repeatedly in the last decades (Asprey and Macchietto, 2000, 2002; Franceschini and Macchietto, 2008; Körkel et al., 2004; Pukelsheim, 1993), applications to ill-posed problems can also be found (Bardow, 2008; Bitterlich and Knabner, 2003; Haber et al., 2008, 2010; Horesh et al., 2010; Lahmer, 2011; O’Sullivan, 1986). However, ill-posed problems have to be handled carefully as they yield a biased estimator. To deal with this, some researchers (Haber et al., 2010; Horesh et al., 2010; O’Sullivan, 1986) propose an adapted experimental design criterion (i.e., Mean Squared Error, MSE) which considers both, the estimator variance and its bias. This technique is computationally demanding as it leads to a bi-level optimization problem, where at each design iteration the inverse problem must be calculated for quite a large number of models and noise realizations. Applications for the experimental design of several large-scale nonlinear and ill-posed inverse problems using MSE together with Tikhonov regularization have been presented recently, e.g., a borehole tomography problem in Horesh et al. (2010), geophysical electromagnetic experiments in Haber et al. (2010) and damage identification problem of gravity dams in Lahmer, 2011.

An important aspect of the OED for nonlinear models is the fact that the design depends on the true parameter values which are unknown (Ford and Silvey, 1980). In the so called sequential approach, experiments are designed, executed and analyzed in a

sequence to indirectly deal with uncertainties in the parameter values (Barz et al., 2010; Doví et al., 1993; Ford and Silvey, 1980; Körkel et al., 1999). As an extension to this adaptive method an online redesign for dynamic models has been proposed in Mehra (1974) and validated recently in Barz et al. (2013), Galvanin et al. (2009), Jayasankar et al. (2010), Körkel and Arellano-Garcia (2009), Stigter et al. (2006) and Zhu and Huang (2011). Here, especially at the beginning of an adaptive experimental design, when the experimental data is scarce, adequate actions for handling ill-posed PE and OED problems need to be taken to ensure the robustness and reliability of the algorithm.

In this work, an ill-conditioning analysis for local identifiability diagnosis is proposed. It is based on the singular values of the sensitivity matrix and suitable for discrete ill-posed problems in nonlinear PE and OED. It yields a classification of the problem, being either rank-deficient or of ill-determined rank (Hansen, 1998). A link is established between the measures of the presented analysis to the common alphabetic design criteria in OED and information for the solution of ill-posed OED is derived. It is important to point out that this analysis in the context of OED is not yet available in literature. Moreover, different regularization techniques for handling nonlinear ill-posed PE problems from singular value analysis point of view are discussed, namely, orthogonal decomposition based techniques (i.e., SsS and TSVD) (Burth et al., 1999; Golub and Van Loan, 1996; Hansen, 1998; López et al., 2013; Marquardt, 1970; Xu, 1998) and the Tikhonov strategy (Bardow, 2008; Haber et al., 2008, 2010; Hansen, 1998, 2007; Johansen, 1997; Tikhonov and Arsenin, 1977). These techniques are then applied to ill-posed OED problems. To avoid the computational demanding bi-level optimization approach addressed by Haber et al. (2010), Horesh et al. (2010), we only consider the variance contribution of the biased estimator (Bard, 1974; Fessler, 1996). The application of these techniques is presented for three literature case studies with increasing complexity i.e., a semi continuous (fed-batch) fermentation reactor (Asprey and Macchietto, 2002), a reconstruction of biochemical networks (Kremling et al., 2004) and a biological system to water treatment (Cruz Bournazou et al., 2012; Kaelin et al., 2009).

2. Problem statement

We consider dynamic processes described by a system of ordinary differential equations (ODEs):

$$\begin{aligned}\dot{x}(t) &= f(x(t), u, \theta, t) \\ x(t_0) &= x_0\end{aligned}\quad (1)$$

where $t \in [t_0, t_{Nm}] \subseteq \mathbb{R}$ is the independent variable time, $x(t) \in \mathbb{R}^{Nx}$ is vector of dependent state variables, $u(t) \in \mathbb{R}^{Nu}$ the time-varying input action or experimental design variable vector, $\theta \in \mathbb{R}^{Np}$ the unknown parameter vector and initial conditions are given by x_0 .

A vector of observable predicted response variables $y(t) \in \mathbb{R}^{Ny}$ is considered whose elements depend on the solution $x(t) \in \mathbb{R}^{Nx}$ of problem in Eq. (1), see Eq. (2).

$$y(t) = h(x(t)) \quad (2)$$

Additionally, there exists a discrete set of time instances $t_k \in \mathcal{T} = \{t_1, \dots, t_{Nm}\}$ (data sampling time grid) where the measured variables $Y^m = [y^m(t_1)^T, \dots, y^m(t_{Nm})^T]^T \in \mathbb{R}^{Ny \cdot Nm}$ are collected. The corresponding model responses evaluated at $u(t)$ and θ are $Y(u, \theta) = [y(u, \theta, t_1)^T, \dots, y(u, \theta, t_{Nm})^T]^T \in \mathbb{R}^{Ny \cdot Nm}$. The time dependent input actions $u(t)$ can be represented by piece-wise polynomial approximations. Here, a piecewise constant approximation is considered with switching times $t_j \in \mathcal{T}^u = \{t_1, \dots, t_{Nmu}\}$ defining the so called input action time grid. For the sake of simplicity we refer to the corresponding discrete decisions by the vector u .

The true value of the parameter vector is $\theta^* \in \mathbb{R}^{Np}$. All measured discrete data $y_i^m(t_j) \in Y^m$ consists of noisy indirect observations with independent-additive and **normally distributed measurement errors** such that $\varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, see Eq. (3).

$$y_i^m(t_j) = y_i(u, \theta^*, t_j) + \varepsilon_i, \quad \text{with } i = 1, \dots, N_y, \quad j = 1, \dots, N_m \quad (3)$$

The model parameter vector θ is then recovered from the discrete noisy experimental data Y^m by solving the unconstrained PE problem (Bard, 1974).

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta} \Phi^{LSQ}(u, \theta) \\ \Phi^{LSQ}(u, \theta) &= \frac{1}{2} ((Y(u, \theta) - Y^m)^T C_M^{-1} (Y(u, \theta) - Y^m)) \end{aligned} \quad (4)$$

$\hat{\theta} \in \mathbb{R}^{Np}$ is the asymptotically unbiased estimator containing the best currently available estimate of the true parameter vector θ^* , if the PE is well-posed. The objective function Φ^{LSQ} denotes the weighted nonlinear least-squares criterion.

The **weighting matrix** $C_M^{-1} \in \mathbb{R}^{N_y \cdot N_m \times N_y \cdot N_m}$ is the inverse of the variance-covariance matrix C_M of the measurement errors in the data. C_M is assumed to be a diagonal matrix with the variance σ_i^2 of each measurement i in its diagonal entries. $C_M = \Sigma_y \Sigma_y^T$ is defined such that $\Sigma_y \in \mathbb{R}^{N_y \cdot N_m \times N_y \cdot N_m}$ is the **measurement standard deviation matrix** whose i th diagonal element is given by σ_i . With the previous assumptions the solution of Eq. (4) is equivalent to the maximum likelihood (Bard, 1974).

For nonlinear PE the solution of Eq. (4) is iteratively solved according to Eq. (5) (Bard, 1974).

$$\hat{\theta}_{k+1} = \hat{\theta}_k + \rho_k v_k \quad (5)$$

In Eq. (5), ρ_k is the step size and v_k is the step direction in the k th iteration. Typical gradient-based solution methods are steepest-descent and Newton type as Gauss–Newton, Levenberg–Marquardt (Marquardt, 1963) and trust region methods (Sorensen, 1982).

In those approaches, the Jacobian ($J_\theta(u, \theta) \in \mathbb{R}^{Np \times 1}$) and the Hessian ($H_\theta(u, \theta) \in \mathbb{R}^{Np \times Np}$) matrices (Eqs. (6) and (7)) of Φ^{LSQ} in Eq. (4) are required to compute the step direction v_k . For instance in Newton's method $v_k = R_k q_k$ where $R_k = H_\theta^{-1}$ and $q_k = J_\theta$ (Bard, 1974).

$$J_\theta(u, \theta) = \nabla_\theta \Phi^{LSQ} = dY(u, \theta)/d\theta^T C_M^{-1} (Y(u, \theta) - Y^m) \quad (6)$$

$$\begin{aligned} H_\theta(u, \theta) &= \nabla_\theta^2 \Phi^{LSQ} = dY(u, \theta)/d\theta^T C_M^{-1} dY(u, \theta)/d\theta \\ &+ d^2 Y(u, \theta)/d\theta^2 C_M^{-1} (Y(u, \theta) - Y^m) \end{aligned} \quad (7)$$

The Jacobian matrix $J_\theta(u, \theta)$ in Eq. (6) is based on the matrix $S(u, \theta) = dY(u, \theta)/d\theta \in \mathbb{R}^{N_y \cdot N_m \times Np}$ and the residual vector $Z = (Y(u, \theta) - Y^m)$. The matrix $S(u, \theta)$ is the sensitivity matrix of the residual vector $Z \in \mathbb{R}^{N_y \cdot N_m \times 1}$. Following the Gauss approximation (i.e., neglecting the second term in Eq. (7)) and using $S(u, \theta)$ the Hessian matrix $H_\theta(u, \theta)$ can be represented by Eq. (8).

$$\begin{aligned} H_\theta(u, \theta) &\approx S(u, \theta)^T C_M^{-1} S(u, \theta) = (\Sigma_y^{-1} S(u, \theta))^T (\Sigma_y^{-1} S(u, \theta)) \\ &= \tilde{S}(u, \theta)^T \tilde{S}(u, \theta) \end{aligned} \quad (8)$$

H_θ is the $Np \times Np$ dispersion matrix, which **is identical to the Fisher information matrix (FIM)** and belongs to the subspace of positive semi-definite matrices $PSD(Np)$ (Pukelsheim, 1993).¹ Note that in Eq. (8), C_M is disjointed by the **measurement standard**

deviation matrix Σ_y and accordingly the **scaled sensitivity matrix** $\tilde{S} = \Sigma_y^{-1} S$ is defined.

2.1. Identifiability problems and ill-conditioning analysis

Typically, identifiability problems in PE are detected based on the analysis of H_θ (see Eq. (8)), and the observation, that H_θ is: a) singular² or b) “almost” singular from a numerical point of view (Asprey and Macchietto, 2000; Franceschini and Macchietto, 2008; Vajda et al., 1989; Walter and Pronzato, 1996). In the former, H_θ has exact linear dependence among its columns (a case rarely encountered in actual practice) and in the latter, its columns have nearly linear dependencies (Belsley et al., 1980). Having so, H_θ is either not invertible or invertible but can numerically run into problems (ill-conditioning). The estimation problem is thus exactly or nearly singular. In those cases the assumption, that H_θ is of full rank and belongs to the set of positive definite matrices $PD(Np)$ (Bard, 1974) may not hold.³ Indefinite Hessians can cause unacceptable (inflated) search directions (see Eq. (5)) (Marquardt, 1970). Moreover, for problems where H_θ is ill-conditioned (nearly singular problems), a large variance (uncertainty) in some sub-space of the parameter space can be found (Bard, 1974; Vajda et al., 1989; Johansen, 1997). Estimates may be obtained far away from the true parameter values and thus some parameters cannot be accurately estimated and are non-identifiable (Vajda et al., 1989).

It has to be noted, that well established solution methods as Levenberg–Marquardt (Marquardt, 1963, 1970) and trust region (Moré and Sorensen, 1983; Sorensen, 1982) are adapted to handle indefinite matrices (Bard, 1974; Kaltenbacher et al., 2008). However, for nonlinear PE problems with highly ill-conditioned matrices, Levenberg–Marquardt method exhibits reduced convergence rates (Kaltenbacher et al., 2008) and for trust-region methods efficiency and stability are negatively affected. Moreover, Levenberg–Marquardt method is based on the Tikhonov regularization of the linearized problem in Newton's method and trust-region method is a modification of Newton's method with origin from Levenberg–Marquardt method.

The eigensystem (eigenvalues and eigenvectors) of H_θ has been used to analyze local identifiability problems (Brun et al., 2002; Burth et al., 1999; Vajda et al., 1989; Vélez-Reyes, 1992; Vélez-Reyes and Verghese, 1995). Here, the presence of a “small” eigenvalue of H_θ is a discernible proof of identifiability problems. Indeed, a “small” eigenvalue of H_θ evidences its ill-conditioning (Belsley et al., 1980). Nonetheless, in cases where H_θ is extremely ill-conditioned the computation of the eigensystem has weak numerical stability (Belsley et al., 1980).

At this point, it is useful to return to the intimate connection of H_θ and $\tilde{S}$ in Eq. (8). It is obvious that if H_θ is ill-conditioned, $\tilde{S}$ is also ill-conditioned i.e., it does not have full rank. This happens, for example, for columns of $\tilde{S}$ with small values (even close to or equal zero) due to insensitive parameters or nearly identical columns or columns which are linear combinations of others because of correlated parameters. Differently expressed, if $\tilde{S}$ is ill-conditioned then H_θ is ill-conditioned, the PE is ill-posed and at least has local identifiability problems. To establish the ill-conditioning analysis for identifiability diagnosis based on $\tilde{S}$, in the sequel some important definitions are introduced first: singular value decomposition (SVD), condition number, collinearity index and numerical ϵ -rank.

² H_θ is singular if and only if H_θ is not invertible, or $\det(H_\theta) = 0$, or H_θ is not of full rank, or $\lambda_{\min}(H_\theta) = 0$.

³ $PD(Np)$ is the subspace of the positive definite matrices in the space $Sym(Np)$. If $H_\theta \in PD(Np)$ its smallest eigenvalue $\lambda_{\min}(H_\theta) > 0$ is greater than zero (Pukelsheim, 1993).

¹ $PSD(Np)$ is the subspace of the positive semi-definite matrices in the space $\mathbb{R}^{Np \times Np}$ of all symmetric matrices $Sym(Np)$. If $H_\theta \in PSD(Np)$, its smallest eigenvalue $\lambda_{\min}(H_\theta)$ is greater than or equal zero (Pukelsheim, 1993).

Moreover, the relation of the eigensystem of H_θ and the SVD of $\tilde{S}$ will be analyzed.

2.1.1. Singular value decomposition (SVD)

The analysis of the ill-conditioning of $\tilde{S}$ is done using the orthogonal decomposition or rank-revealing SVD shown in Eq. (9) (Belsley et al., 1980; Hansen, 1998; Jacobson, 1985).

$$\tilde{S} = U S_\nu V^T = \sum_{i=1}^{N_p} u_i \varsigma_i v_i^T \quad (9)$$

In Eq. (9), $U \in \mathbb{R}^{N_y \cdot N_m \times N_y \cdot N_m}$ and $V \in \mathbb{R}^{N_p \times N_p}$ are real **orthogonal** matrices (e.g., $U^T U = I_{N_y \cdot N_m}$) and $S_\nu = \text{diag}(\varsigma_1, \dots, \varsigma_{N_p})$, with ς_i being the i th singular value of $\tilde{S}$ (ς_i ordered according to magnitude as $\varsigma_1 \geq \varsigma_2 \geq \dots \geq \varsigma_{N_p} \geq 0$). Hereinafter the set of all singular values of $\tilde{S}$ is referred to as the singular value spectrum (SVs). The vectors $u_i \in \mathbb{R}^{N_y \cdot N_m \times 1}$ and $v_i \in \mathbb{R}^{N_p \times 1}$ are the left and right singular vectors of $\tilde{S}$.

Based on the SVD in Eq. (9) and the properties of orthogonal and diagonal matrices, H_θ in Eq. (8) is written as shown in Eq. (10).

$$H_\theta = V S_\nu^2 V^T = \sum_{i=1}^{N_p} v_i \varsigma_i^2 v_i^T \quad (10)$$

It is worth noting that Eq. (10) is the eigensystem of $H_\theta = \tilde{S}^T \tilde{S}$ whose diagonal elements of **matrix S_ν^2 (the square of the singular value ς_i)** are the eigenvalues of the real symmetric matrix H_θ . This relation is shown in Eq. (11), where the eigenvalues of H_θ are denoted as $\lambda_i(H_\theta)$, with $i = 1, \dots, N_p$.

$$\varsigma_i^2 = \lambda_i(H_\theta) \quad (11)$$

The SVD of $\tilde{S}$ provides information that encompasses the information given by the eigensystem of H_θ . As a practical matter, there are reasons for preferring the use of the SVD (Belsley et al., 1980). Firstly, it applies directly to the sensitivity matrix $\tilde{S}$ instead of H_θ which is also used for local identifiability analysis of nonlinear PE.⁴ Secondly, although the computation of the SVD is typically more expensive than the eigensystem, in the context of larger problems with sparse matrices the SVD can be performed more efficiently (Bingham and Mannila, 2001; Wu et al., 1997). Moreover, the computation of $H_\theta = \tilde{S}^T \tilde{S}$, corresponding to $(N_y \cdot N_m) \cdot N_p^2$ operations, as well as the supplementary N_p^3 operations for the construction of the inverse of H_θ (see Eq. (17)) can be saved. Thirdly, the SVD of $\tilde{S}$, where it is ill-conditioned, can be computed with much greater numerical stability than the eigensystem of H_θ (Belsley et al., 1980). One reason is that the condition number from H_θ is the square of the condition number from $\tilde{S}$. In this context, when the system is extremely ill-conditioned the precision of the smallest eigenvalues of H_θ (FIM) is really doubtful (Belsley et al., 1980). For instance, additional numerical errors could take place in the computation of H_θ in Eq. (8) which could lead to the presence of unexpected negative eigenvalues (see Section 5.1.3 for application problem E3). Therefore, the corresponding eigenvalues of the parameter variance-covariance matrix are also inaccurate which significantly impedes the computation of OED (see Section 2.4).

2.1.2. Condition number (κ) and collinearity index (γ)

There is one condition for defining a matrix as ill-conditioned after applying the foregoing decomposition in Eq. (9), that means a large ratio of the maximum (ς_1) to minimum (ς_{N_p}) singular value. This condition checks how small the minimum singular value ς_{N_p}

is relative to the maximum singular value ς_1 . In some matrices this smallness is defined by comparing ς_{N_p} to the standard zero (Belsley et al., 1980) if ς_1 has moderate value whilst other cases a large ratio might occur for singular values being all far from zero.

The way to measure the mentioned condition is by the condition number $\kappa = \varsigma_1 / \varsigma_{N_p} \geq 1$ (Belsley et al., 1980). When κ is not far from 1, the matrix is said to be well-conditioned, on the contrary when κ is large the matrix is ill-conditioned. **Only a matrix with orthogonal (perfect linearly independent) columns has $\kappa = 1$.** Otherwise, if the matrix has exact linear dependencies among its columns there exists one zero singular value for each one (Belsley et al., 1980; Belsley, 1991), and then $\kappa = \infty$. In most applications of nonlinear PE all singular values of the matrix $\tilde{S}$ are nonzero, because its columns are rarely exactly linearly dependent (except the case that there are some structural non-identifiabilities). Nonetheless, if among the columns of $\tilde{S}$ there exists a near linear dependence (also called collinearity) (Belsley et al., 1980) this will manifest itself in a “small” singular value and then a large κ . **The indicator of the degree of near-linear dependence is the collinearity index $\gamma(\tilde{S}) = 1 / \varsigma_{N_p}$** (Brun et al., 2001, 2002; Grah, 2004). If this quantity is large, $\tilde{S}$ has columns (nearly) linearly dependent, is ill-conditioned and the parameter vector is poorly identifiable. According to Grah (2004) and Brun et al. (2002), empirical upper bounds for the condition number and the collinearity index of the unscaled matrix S are defined by $\kappa^{\max}(S) \cong 1000$ and $\gamma^{\max}(S) \cong 10 - 15$, respectively. The maximum condition number $\kappa^{\max}$ and collinearity index $\gamma^{\max}$ are used to define a well-conditioned sensitivity matrix and a well-posed problem without numerical instabilities due to error propagation and nearly linear dependencies. Accordingly, **$\kappa^{\max}$ assures numerical stability and $\gamma^{\max}$ controls linear dependencies.** Furthermore, because of these thresholds are related to the definition of a “large” ratio in the SVs ($\kappa^{\max}$) and “smallness” of singular values ($\gamma^{\max}$), the user might tune them according to the magnitude of singular values found in the particular application. However the foregoing values are good references.

It is important to note that the **SVD does not have scale invariance**, therefore if a matrix A is scaled by a positive scalar α the corresponding SVs of αA is also scaled by α (i.e., $\alpha A = \sum_{i=1}^{N_p} u_i (\alpha \varsigma_i) v_i^T$, where ς_i and $\alpha \varsigma_i$ are the i th singular values of A and αA , respectively). In the semi-logarithmic graphical representation (used in the sequel), α shifts the spectrum and the new SVs of αA is a parallel curve to the SVs of A . This scaling does not affect the **condition number** because it **is scale invariant**, i.e., $\kappa(\alpha A) = (\alpha \varsigma_1) / (\alpha \varsigma_{N_p}) = \kappa(A)$. **However, it affects the collinearity index**, i.e., $\gamma(\alpha A) = 1 / (\alpha \varsigma_{N_p}) = \gamma(A) / \alpha$. For the scaled sensitivity matrix $\tilde{S}$ in Eq. (8) if the standard deviation of the measurement errors is constant (i.e., $\sigma_i = \sigma$ for $i = 1, \dots, N_y \cdot N_m$), according to the previous analysis $\alpha = 1/\sigma$ and $\gamma(\tilde{S}) = \sigma \gamma(S)$ and $\kappa(\tilde{S}) = \kappa(S)$. The same applies for $\kappa^{\max}(\tilde{S})$ and $\gamma^{\max}(\tilde{S})$. The scale-invariance catalogs the **condition number as a robust indicator for ill-conditioning.**

2.1.3. Numerical ϵ -rank r_ϵ and classification of discrete ill-posed problems

The next important definition to be set is the **numerical ϵ -rank r_ϵ of $\tilde{S}$ which is the size of the largest collection of linearly independent columns of $\tilde{S}$.** In terms of singular values, having a well-determined gap between large and small singular values limited by a ϵ -threshold, the number of these largest singular values ($\varsigma_1, \dots, \varsigma_{r_\epsilon}$) reveals the numerical ϵ -rank r_ϵ of the matrix (see Eq. (12)).

$$\varsigma_1, \dots, \varsigma_{r_\epsilon} \geq \epsilon > \varsigma_{r_\epsilon+1}, \dots, \varsigma_{N_p} \quad (12)$$

It is obvious, that only a well-conditioned matrix $\tilde{S}$ is of full-rank (without the presence of small singular values). Otherwise, the corresponding problem is considered to be rank-deficient (Hansen,

⁴ If a model is locally unidentifiable at θ , then the columns of the sensitivity matrix $\tilde{S}$ are linearly dependent, and $\tilde{S}^T \tilde{S}$ is singular (Jacobson, 1985; Vajda et al., 1989).

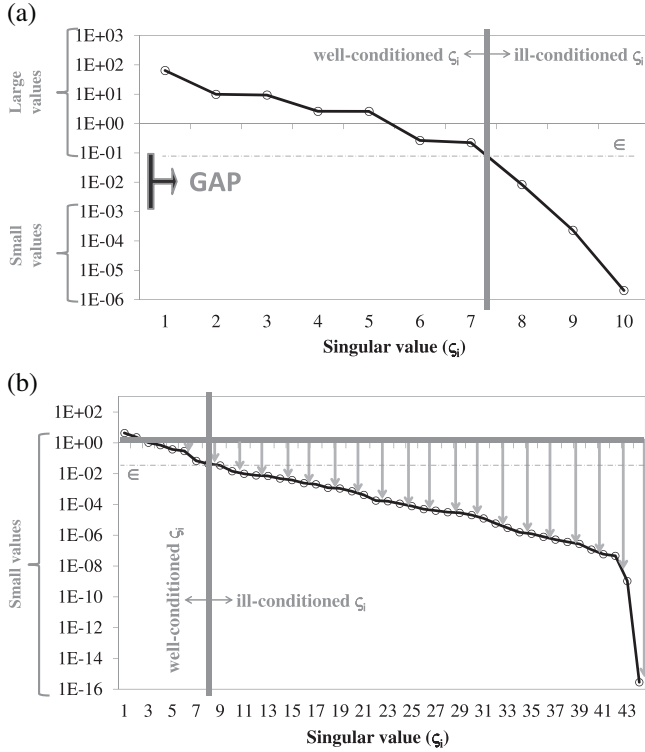


Fig. 1. Singular value spectrum (SVs) of ill-posed problems: (a) rank-deficient and (b) ill-determined rank problems.

1998, 2007). Moreover, problems with a SVs decaying gradually to zero without a gap between singular values belong to the ill-determined rank problem class (Hansen, 1998, 2007).

The lower bound on the SVs, namely the ϵ -threshold, is calculated from the maximum condition number $\kappa^{\max}(\tilde{S})$ and collinearity index $\gamma^{\max}(\tilde{S})$ defined in Section 2.1.2, through Eq. (13).

$$\epsilon = \max \left\{ \epsilon_k = \frac{\zeta_1}{\kappa^{\max}(\tilde{S})}, \quad \epsilon_\gamma = \frac{1}{\gamma^{\max}(\tilde{S})} \right\} \quad (13)$$

In Fig. 1, SVs for both, rank-deficient and ill-determined rank problems are depicted. For the rank-deficient problem in Fig. 1a, the numerical ϵ -rank (i.e., number of linear independent columns of $\tilde{S}$) can be well-determined by using the ϵ -threshold. Rank-deficient problems have either a large condition number $\kappa(\tilde{S})$ and/or a large collinearity index $\gamma(\tilde{S})$. Furthermore, the largest singular value ζ_1 is larger than 1.

For the ill-determined rank problem in Fig. 1b, the dominant feature is the presence of singular values close to zero. Both, large values of $\gamma(\tilde{S})$ and $\kappa(\tilde{S})$ are to be expected. Moreover, almost the complete SVs is below 1 and even the largest singular value ζ_1 is small (see Fig. 1b). Although, ill-determined rank problems do not possess a well-defined r_ϵ , the ϵ -threshold can still be used to establish a new well-conditioned problem (with only well-conditioned singular values) which is related to the amount of available linearly independent information.

As mentioned for the condition number, the rank of a matrix A is also scale invariant. However, when applying the collinearity index to the scaled matrix αA , the ϵ -threshold for computing r_ϵ must be modified as $\alpha \epsilon = \max \{ \alpha \epsilon_{\kappa(A)}, \alpha \epsilon_{\gamma(A)} \}$ such that $r_{\alpha \epsilon}(\alpha A) = r_\epsilon(A)$.⁵

⁵ For the scaled sensitivity matrix $\tilde{S} = S/\sigma$ considering $\alpha = 1/\sigma$ its maximum collinearity index should be also scaled as $\gamma^{\max}(\tilde{S}) = \gamma^{\max}(S)/\alpha$. Doing so, $\epsilon_\gamma(\tilde{S}) = 1/\gamma^{\max}(\tilde{S}) = \epsilon_\gamma(S)/\sigma$ with $\epsilon_\gamma(S) = 1/\gamma^{\max}(S)$ and then $\epsilon(\tilde{S}) = \epsilon(\alpha S) = \max\{\epsilon_\kappa(S)/\sigma, \epsilon_\gamma(S)/\sigma\} = 1/\sigma \epsilon(S) = \alpha \epsilon(S)$.

This fact shows that the ill-conditioning of a problem is not affected by this scaling but indicators based on the singular values (and also eigenvalues) need to be applied carefully. On the other hand, it is recommended for all computations (PE, OED and ill-conditioning analysis) to normalize predicted response variables and parameters as $\tilde{y}_i(u, \hat{\theta}, t_k) = y_i(u, \hat{\theta}, t_k) / \max_{t_k \in \mathcal{T}} \{y_i^m(t_k)\}$ with $i = 1, \dots, N_y$ and $\tilde{\theta}_j = \hat{\theta}_j / \theta_j^{IG}$ with $j = 1, \dots, N_p$, respectively. Doing so, the estimation problem is normalized and the matrix $\tilde{S}$ is considered in its standard form. This normalized matrix $\tilde{S}$ preserves all information of the system (Buzzi-Ferraris, 2011). A standardized matrix $\tilde{S}$ has reduced influence of parameters and predicted response variables with different orders of magnitude which can mask the ill-conditioning analysis and identifiability diagnosis.⁶

2.1.4. Guideline for ill-conditioning analysis to diagnose identifiability problems

The presented guideline is based on an ill-conditioning analysis of the sensitivity matrix and can be applied to detect practical identifiability problems for nonlinear discrete ill-posed problems. It is based on a procedure for conditioning diagnostics in linear estimation presented by Belsley et al. (1980) and Belsley (1991). This procedure is here modified for nonlinear models and displayed in Fig. 2. The main steps are: compute the condition number (related to numerical stability) and collinearity index (related to parameter linear interdependencies); determine well-conditioned singular values (related to the numerical rank); study the influence of ill-conditioned singular values on the parameter variance; and select preliminary candidates for unidentifiable parameters.

A step regarding the classification of ill-posedness is included. This classification leads to a recommendation of an appropriate regularization technique either for PE or OED. If the problem is rank-deficient, available linearly independent information should be exploited by applying either SsS (Section 2.2.1) or TSVD (Section 2.2.2). For ill-determined problems the common practice is to apply Tikhonov regularization (Section 2.2.3). However, each mentioned technique can be used for both classes of problems.

The procedure in Fig. 2 consists as follows:

0. Compute scaled sensitivity matrix $\tilde{S}$
1. Conduct a SVD of $\tilde{S}$ according to Eq. (9) and get the SVs.
2. Compute the condition number $\kappa(\tilde{S})$ and collinearity index $\gamma(\tilde{S})$ (see Section 2.1.2).
3. Check if $\kappa(\tilde{S}) \leq \kappa^{\max}(\tilde{S})$, Λ , $\gamma(\tilde{S}) \leq \gamma^{\max}(\tilde{S})$. If this is true, $\tilde{S}$ is well-conditioned and the problem is well-posed. The PE or OED can be solved without regularization. No further analysis is necessary. If it is false, $\tilde{S}$ is ill-conditioned (identifiability problems exist), the problem is ill-posed and a regularization is required.
4. Analyze the ill-conditioning of $\tilde{S}$ (i.e., rank-deficient or ill-determined rank class) using its SVs, according to Fig. 1 (see Section 2.1.3) and classify the problem. A rank-deficient problem is considered a mild ill-posed problem, whilst an ill-determined rank problem is a severe case of ill-posedness.
5. Set a problem-specific ϵ -threshold (Eq. (13)) to assess the relative smallness of each singular value with respect to the largest singular value (related to $\kappa^{\max}(\tilde{S})$) and to zero (related to $\gamma^{\max}(\tilde{S})$).
6. Cut the SVs at the ϵ -threshold.
7. Choose the well-conditioned singular values of the SVs such that the singular values ζ_i for $i = 1, \dots, N_p$, are nominated well-conditioned⁷ if $\zeta_i \geq \epsilon$. Otherwise, ζ_i is ill-conditioned. The

⁶ In linear estimation one recommended column equilibration is to scale the columns of $\tilde{S}$ to unit length (Belsley, 1991).

⁷ The nomination of a singular value as either well or ill-conditioned is related to its potential to promote either a well or ill-conditioned sensitivity matrix.

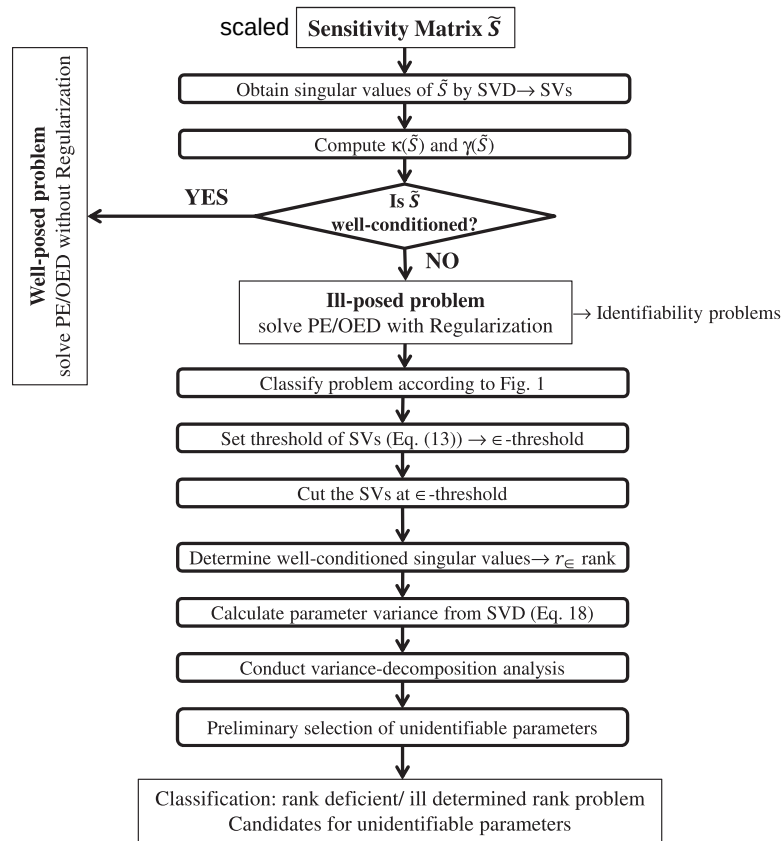


Fig. 2. Guideline for the analysis of nonlinear ill-posed problems to detect practical identifiability problems.

number of well-conditioned singular values defines the numerical ϵ -rank, r_ϵ (see Section 2.1.3).

8. Calculate the **parameter variance-covariance matrix from SVD** according to Eq. (17) (see Section 2.3). Select the parameter variance.
9. Conduct a **parameter variance-decomposition analysis** and compute the variance-decomposition proportion π_i for $i = 1, \dots, N_p$ (see Section 2.3). This decomposition determines the influence of each ill-conditioned singular value on the variance of each parameter.
10. Do a preliminary selection of practically unidentifiable parameter candidates. Choose those parameters with large π_i (say, greater than 0.5) as they are associated to the ill-conditioned singular values. If many ill-conditioned singular values substantially contribute to the variance of two or more parameters and the sum of their values π_i exceeds 0.5 (even if neither of the individual values is greater than 0.5), these two or more parameters are involved in at least one competing near dependence and are unidentifiable parameter candidates.

The foregoing guideline correctly diagnoses identifiability problems and quantifies the dimension of the identifiable parameter subset (i.e., r_ϵ). However, the selection of practically unidentifiable parameters is a rough procedure. Therefore, a more elaborated analysis should be applied to determine the identifiable parameters (for instance, SsS in Section 2.2.1, or see also Chu and Hahn (2007), Chu and Hahn (2012) and Kravaris et al. (2013) for more advanced methods).

2.2. Numerical regularization for PE

Ill-posedness can be handled by incorporating a priori information about the desired solution via the use of transformations,

which stabilize the problem and lead the solution to a useful and stable region which will be hopefully near the desired (Kaltenbacher et al., 2008; Hansen, 1998, 2007). When a regularization technique (Reg) is applied to Eq. (4) the solution norm, either 2-norm (i.e., $\|\hat{\theta}^{\text{Reg}}\|_2$) or an appropriate seminorm, and the residual value (i.e., $Z = (Y(u, \hat{\theta}^{\text{Reg}}) - Y^m)$) should remain small. Nevertheless, the regularized solution $\hat{\theta}^{\text{Reg}}$ is obtained from a modified problem. Hence $\hat{\theta}^{\text{Reg}}$ is generally biased but also has a reduced variance as the regularization decreases the size of the solution's covariance matrix (Fessler, 1996; Hansen, 1998, 2007; Johansen, 1997).

For rank-deficient problems with matrix $\tilde{S}$ whose r_ϵ exists, regularization strategies which exploit the filtering of a subset either of the parameters or the singular values can be applied. The key feature of these strategies is to extract the linearly independent information of $\tilde{S}$ and to transform the problem into a well-conditioned one (Hansen, 1998, 2007).

Ill-determined rank problems (i.e., r_ϵ of $\tilde{S}$ cannot exactly be determined independently of the value of ϵ used) could generally be treated with strategies for rank-deficient problems. However, better suited are regularization techniques which introduce a penalty term to ensure smoothness of the solution, for instance the Tikhonov regularization (known in linear cases as ridge regression) (Hansen, 1998, 2007; Johansen, 1997; Marquardt, 1970). Here the ill-conditioning of $\tilde{S}$ is also directly modified. Instead of selecting its well-conditioned information (maintaining the formulation in Eq. (4)), a new well-conditioned matrix (i.e., the regularization parameter multiply by a predefined full row rank matrix) is added to $\tilde{S}$ (Hansen, 1998, 2007; Hoerl and Kennard, 1970a, 1970b; Marquardt, 1970).

The three regularization techniques, namely parameter Subset Selection (SsS), Truncated Singular Value Decomposition (TSVD),

and Tikhonov regularization (Tikh) are studied in the sequel. The choice of these techniques lies on their broad application, good performance and easy implementation. Moreover, SsS and TSVD have a close relationship to the here referred ill-conditioning analysis (see Section 2.1.4) because they are also based on the singular value analysis. It is important to point out, that all mentioned regularization techniques are aimed at getting rid of the ill-conditioning of the original problem and turning it into a new well-posed problem (Hansen, 1998; Kravaris et al., 2013). In the field of parameter subset selection, more sophisticated methods have been proposed recently which are also based on orthogonalization methods (Chu and Hahn, 2007, 2012; Wu et al., 2011). The main differences are in the chosen orthogonalization algorithm (e.g., Gram–Schmidt or householder transformation) and how parameter combinations are selected for estimation (possible criteria are, e.g., maximum determinant of reduced FIM, largest column norm of reduced sensitivity matrix or lowest MSE). In Chu and Hahn (2012) the selection is formulated as an optimization problem. However, all strategies coincide in finding a parameter subset which generates a well-conditioned sensitivity matrix in the reduced parameter space.

Furthermore, each technique uses a regularization parameter which must be tuned according to the gravity of the ill-conditioning in the particular application. The sensitivity matrix $\tilde{S}^{\text{Reg}}$ with $\text{Reg} = \{\text{SsS}, \text{TSVD}, \text{Tikh}\}$ is computed and used to obtain H_{θ}^{Reg} (see Eq. (8)). The inverse of H_{θ}^{Reg} (more stable) is not only the basis for the calculation of the step direction v_k in Eq. (5) but also for the construction of the parameter variance-covariance matrix (Eq. (17)), see Section 2.3.

2.2.1. Parameter subset selection (SsS)

The idea of dealing with ill-conditioned PE by selecting a well-conditioned parameter subset and solving a new (but reduced) PE has been applied first in linear least squares estimation with non-orthogonal data (Hoerl and Kennard, 1970b) and nonlinear PE with inflated step directions (v_k in Eq. (5)) (Marquardt, 1970). More recently, systematic strategies for parameter set selection have been proposed (see for instance Brun et al., 2001; Burth et al., 1999; Chandrakant and Bisaria, 1998; Chu and Hahn, 2007, 2012; Golub and Van Loan, 1996; Hansen, 1998; McLean et al., 2012; López et al., 2013; Vélez-Reyes, 1992; Vélez-Reyes and Verghese, 1995; Wu et al., 2011; Yao et al., 2003).

In this work an algorithm for local identifiability analysis is used, which has been presented by Golub and Van Loan (1996) and Vélez-Reyes (1992) and Vélez-Reyes and Verghese (1995) with applications to linear and nonlinear models, respectively. The original algorithm is based on orthogonal projections of the Hessian matrix H_{θ} (see Eq. (8)) and is one of the most widely used among several approaches for parameter subset selection (Kravaris et al., 2013). It was modified by López et al. (2013) using the sensitivity matrix S .

The dimension of the identifiable parameter subset corresponds to r_{ϵ} of $\tilde{S}$, see Section 2.1.3. In addition, the analyzed parameter

vector $\hat{\theta}$ is reordered according to the linear independence of the columns of $\tilde{S}$ by applying QRP decomposition (via Householder transformation). This decomposition reads: $\tilde{S}\Pi = QR$, where $Q \in \mathbb{R}^{N_y \cdot N_m \times N_y \cdot N_m}$ is an orthogonal matrix, $R \in \mathbb{R}^{N_y \cdot N_m \times N_p}$ is an upper-triangular matrix with decreasing diagonal elements and $\Pi \in \mathbb{R}^{N_p \times N_p}$ is a permutation matrix. Π orders the columns of $\tilde{S}$ according to linear independence. This means that the first columns of $\tilde{S}\Pi$ are the largest independent set of columns of $\tilde{S}$. The reordered parameter vector reads $\tilde{\theta} = \Pi^T \hat{\theta}$ (Burth et al., 1999; Golub and Van Loan, 1996). The result is $\tilde{\theta} = [\tilde{\theta}(r_{\epsilon})^T, \tilde{\theta}(N_p - r_{\epsilon})^T]^T$. Here, $\tilde{\theta}(r_{\epsilon})$ contains the first r_{ϵ} elements of $\tilde{\theta}$ which are the identifiable parameters, and $\tilde{\theta}(N_p - r_{\epsilon})$ contains the last $N_p - r_{\epsilon}$ elements of $\tilde{\theta}$, which are not identifiable. The linearly independent columns of $\tilde{S}$ (i.e., the sensitivities corresponding to the r_{ϵ} active parameters of $\hat{\theta}$) are stored in the reduced sensitivity matrix $\tilde{S}^{\text{SsS}} \in \mathbb{R}^{N_y \cdot N_m \times r_{\epsilon}}$. $\tilde{S}^{\text{SsS}}$ is well-conditioned and has full-column-rank because of $\kappa(\tilde{S}^{\text{SsS}}) \leq \kappa^{\max}(\tilde{S})$ and $\gamma(\tilde{S}^{\text{SsS}}) \leq \gamma^{\max}(\tilde{S})$ (López et al., 2013).

For the solution of nonlinear problems this local regularization technique should be applied repeatedly during the search. The status of parameters (being active or nonactive) is then iteratively refined and nonactive parameters ($\tilde{\theta}(N_p - r_{\epsilon})$) are fixed at prior estimates ($\tilde{\theta}^{(N_p - r_{\epsilon})}$) (López et al., 2013). The corresponding reduced PE problem (Eq. (14)) reads:

$$\begin{aligned} \hat{\theta}^{(r_{\epsilon})} &= \arg \min_{\tilde{\theta}(r_{\epsilon})} \Phi^{\text{SsS}}(u, \tilde{\theta}(r_{\epsilon}), \tilde{\theta}(N_p - r_{\epsilon})) \\ \Phi^{\text{SsS}}(u, \tilde{\theta}(r_{\epsilon}), \tilde{\theta}(N_p - r_{\epsilon})) &= \Phi^{\text{LSQ}}(u, \tilde{\theta}(r_{\epsilon}), \tilde{\theta}(N_p - r_{\epsilon})) \end{aligned} \quad (14)$$

The regularized Jacobian $J_{\theta}^{\text{SsS}} \in \mathbb{R}^{N_y \cdot N_m \times r_{\epsilon}}$ and Hessian $H_{\theta}^{\text{SsS}} \in \mathbb{R}^{r_{\epsilon} \times r_{\epsilon}}$ matrices of the objective in Eq. (14) are given in Table 1. They are obtained from the regularized sensitivity matrix $\tilde{S}^{\text{SsS}} \in \mathbb{R}^{N_y \cdot N_m \times r_{\epsilon}}$. Note that $\tilde{S}^{\text{SsS}} \subseteq \tilde{S}$ and $H_{\theta}^{\text{SsS}} \subseteq H_{\theta}$ is computed according to the assumptions in Eq. (8). Furthermore, according to the interlacing inequality of the singular values, each i th singular value of the sub-matrix $\tilde{S}^{\text{SsS}}$ is at most as large as the corresponding i th singular value of the original/complete matrix $\tilde{S}$, i.e., $\varsigma_i(\tilde{S}^{\text{SsS}}) \leq \varsigma_i(\tilde{S})$ (Thompson, 1972; Carlson, 1983). The regularization parameter in this technique is the ϵ -threshold (see Eq. (13) for its determination) which is involved in the computation of r_{ϵ} .

2.2.2. Truncated singular value decomposition (TSVD)

In TSVD a new problem with a well-conditioned but rank-deficient sensitivity matrix $\tilde{S}^{\text{TSVD}} \in \mathbb{R}^{N_y \cdot N_m \times N_p}$ is derived (Hansen, 1998, 2007; Xu, 1998). The matrix $\tilde{S}^{\text{TSVD}}$ is obtained by truncating the SVs of $\tilde{S}$ at r_{ϵ} and substituting the small nonzero singular values $\varsigma_{r_{\epsilon}+1}, \dots, \varsigma_{N_p}$ with exact zeros (see Table 1). It should be noted that the same truncation principle is used to compute the generalized inverse of $A = X^T X$ when the existence of a distribution of eigenvalues near zero complicates the calculus of its inverse (Marquardt, 1970). In TSVD, the ϵ -rank r_{ϵ} is computed as described in Section 2.1.3. Thus, the regularization parameter in this technique is the ϵ -threshold (see Eq. (13) for its determination). The matrix

Table 1

Definition of the specific derivatives used for “SsS”, “TSVD” and Tikhonov (“Tikh”) regularization; regularization (“Reg”) equal “None” refers to the original PE problem.

Reg	$\tilde{S}^{\text{Reg}}$	J_{θ}^{Reg}	H_{θ}^{Reg}	$\frac{\partial^2 \Phi^{\text{Reg}}}{\partial u \partial y m}$	C^{Reg}
None	$\Sigma_y^{-1} \partial Y / \partial \theta$	$\tilde{S}^T Z$	$\tilde{S}^T S$	$-\tilde{S}^T$	$[H_{\theta}]^{-1}$
SsS	$\Sigma_y^{-1} \partial Y / \partial \tilde{\theta}(r_{\epsilon})$	$\tilde{S}^{\text{SsS}T} Z$	$\tilde{S}^{\text{SsS}T} \tilde{S}^{\text{SsS}}$	$-\tilde{S}^{\text{SsS}T}$	$[H_{\theta}^{\text{SsS}}]^{-1}$
TSVD	$\sum_{i=1}^{r_{\epsilon}} u_i \varsigma_i v_i^T$	$\tilde{S}^{\text{TSVD}T} Z$	$\left[\sum_{i=1}^{r_{\epsilon}} v_i \varsigma_i^2 v_i^T \right]$	$-\left[\sum_{i=1}^{r_{\epsilon}} v_i \varsigma_i v_i^T \right]$	$\left[\sum_{i=1}^{r_{\epsilon}} \frac{v_i v_i^T}{\varsigma_i^2} \right]$
Tikh	$\begin{bmatrix} \tilde{S}^{\text{None}} \\ \lambda L \end{bmatrix}$	$J_{\theta}^{\text{None}} + \frac{1}{2} \lambda^2 J_{\Omega(\theta)}$	$H_{\theta}^{\text{None}} + \frac{1}{2} \lambda^2 H_{\Omega(\theta)}$	$-\tilde{S}^{\text{None}T}$	$[H_{\theta}^{\text{Tikh}}]^{-1}$

$\tilde{S}^{TSVD}$ is rank-deficient. It is the closest rank- r_ϵ approximation to $\tilde{S}$ with residual 2-norm given by $\|\tilde{S} - \tilde{S}^{TSVD}\|_2 = \varsigma_{r_\epsilon+1}$ (Hansen, 1998, 2007).

By applying the TSVD technique the parameter space dimension and the objective of the original PE problem in Eq. (4) remains unchanged, see Eq. (15).

$$\hat{\theta}^{TSVD} = \arg \min_{\theta} \Phi^{TSVD}(u, \theta) \quad (15)$$

$$\Phi^{TSVD}(u, \theta) = \Phi^{LSQ}(u, \theta)$$

The regularized Jacobian $J_{\theta}^{TSVD} \in \mathbb{R}^{Ny \times Nm \times Np}$, and Hessian $H_{\theta}^{TSVD} \in \mathbb{R}^{Np \times Np}$ matrices are calculated based on the regularized sensitivity matrix $\tilde{S}^{TSVD}$ according to Eqs. (6) and (7), respectively, see Table 1.

It has to be noted, that as for most PE problems the parameters are somehow correlated, all parameter sensitivities are affected by the truncation of one or several singular values of $\tilde{S}$.

2.2.3. Tikhonov regularization

This regularization modifies the identification problem in Eq. (4) by incorporating assumptions about the size and smoothness of the desired solution (Hansen, 1998, 2007; Johansen, 1997; Tikhonov and Arsenin, 1977). This a priori information is considered as additional penalty (regularizing) term $\Omega(\theta)$ in the objective function (see Eq. (16)). The purpose is to attract non-identifiable parameters toward their corresponding values in a predefined vector $\theta^R \in \mathbb{R}^{Np \times 1}$ (Johansen, 1997). Indeed, the ill-conditioning of $\tilde{S}$ is eliminated by introducing a new well-conditioned matrix $\tilde{S}^{Tikh}$ of full rank (see Table 1).

$$\hat{\theta}^{Tikh} = \arg \min_{\theta} \Phi^{Tikh}(u, \theta)$$

$$\Phi^{Tikh}(u, \theta) = \Phi^{LSQ}(u, \theta) + \frac{1}{2} \lambda^2 \Omega(\theta) \quad (16)$$

$$\Omega(\theta) := \|L(\theta - \theta^R)\|_2^2$$

In Eq. (16), the most common choice of the penalty term $\Omega(\theta)$ is the ridge regression stabilizer (Marquardt, 1963; Hoerl and Kennard, 1970a, 1970b). The scalar parameter λ controls the regularization contribution in the objective function Φ^{Tikh} .⁸ The functional $\Omega(\theta)$ is called the discrete smoothing norm and assumed to be twice continuously differentiable. The matrix $L \in \mathbb{R}^{Np \times Np}$ is a diagonal matrix typically the identity matrix $I_{Np \times Np}$. Other approaches define L as a discrete approximation of a derivative operator (e.g., first or second derivative operators) (Hansen, 1998, 2007; Johansen, 1997; Tikhonov and Arsenin, 1977). In this work, $L = I_{Np \times Np}$ and no a priori information, i.e., $\theta^R = 0_{Np \times 1}$ (zero-vector) is considered. The Jacobian $J_{\theta}^{Tikh} \in \mathbb{R}^{Ny \times Nm \times Np}$ and Hessian $H_{\theta}^{Tikh} \in \mathbb{R}^{Np \times Np}$ are given in Table 1. Therein, the first and second derivative of $\Omega(\theta)$ with respect to θ read $J_{\Omega(\theta)} = \partial \Omega(\theta) / \partial \theta = 2\theta$ and $H_{\Omega(\theta)} = \partial^2 \Omega(\theta) / \partial \theta^2 = 2I_{Np \times Np}$, respectively.

2.3. Precision of the parameter estimates

The precision of the unbiased estimate $\hat{\theta}$ in Eq. (4) can be assessed by analysis of the parameter variance-covariance matrix $C(u, \hat{\theta}) \in \mathbb{R}^{Np \times Np}$ (Bard, 1974; Fessler, 1996; Lennart and Caines, 1980; Weijers and Vanrolleghem, 1997). An approximation is usually computed by the inverse of H_{θ} , which is derived from the first-order Taylor expansion, the Cramér-Rao lower bound and the Gauss approximation of H_{θ} in Eq. (8) (Bard, 1974; Fessler, 1996

Lennart and Caines, 1980). Despite the fact that this approximation only approximately applies to nonlinear models, it is widely used (Bard, 1974; Fessler, 1996).

$$C(u, \hat{\theta}) \approx H_{\theta}^{-1}(u, \hat{\theta}) = \sum_{i=1}^{Np} \frac{v_i v_i^T}{\varsigma_i^2} \quad (17)$$

In Eq. (17) the definition of $C(u, \hat{\theta})$ is also given in terms of the SVD of $\tilde{S}$. By using this representation, the variance var of the k th component $\hat{\theta}_k$ can be decomposed as $var(\hat{\theta}_k)$ according to Eq. (18), where $v_{ki} \in v_i$ for $k = 1, \dots, Np$ (Belsley et al., 1980).

$$var(\hat{\theta}_k) = \sum_{i=1}^{Np} v_{ki}^2 / \varsigma_i^2 = \sum_{i=1}^{Np} var_{\varsigma_i}(\hat{\theta}_k) \quad (18)$$

The variance $var(\hat{\theta}_k)$ is the k th diagonal element of $C(u, \hat{\theta})$ and is obtained by a sum of components $var_{\varsigma_i}(\hat{\theta}_k)$ for $i = 1, \dots, Np$, each associated with one and only one of the Np singular values ς_i . The effect of “small” singular values (ill-conditioned singular values) is then an inflated variance of $\hat{\theta}_k$. Accordingly, a system with an ill-conditioned $\tilde{S}$ produces an ill-conditioned $C(u, \hat{\theta})$ which is expressed by inflated parameter variances revealing that the model is practically unidentifiable (Walter, 1987). Parameters with large variance (out of some reasonable given ranges) are considered practically non-identifiable. However, parameter variances highly influenced by the ill-conditioned singular values (selected in Section 2.1.4) should be critically analyzed. For instance, when the problem is severely ill-posed, the selection of practically unidentifiable parameters based on parameter variance can fail (see case study E3 in Section 4.1.3), in this case, all parameters could be assigned as unidentifiable.

Computing the proportion of each variance component $var_{\varsigma_i}(\hat{\theta}_k)$ in Eq. (18) with respect to the complete parameter variance $var(\hat{\theta}_k)$, i.e., $\pi_i = var_{\varsigma_i}(\hat{\theta}_k) / var(\hat{\theta}_k)$, a variance decomposition can be conducted (Belsley et al., 1980). The variance decomposition proportion π_i quantifies the contribution of each singular value ς_i to the variance $var(\hat{\theta}_k)$. The ill-conditioned singular values are expected to have large contribution (say, greater than 0.5 or 50% either as an individual proportion π_i or as a sum of proportions) to parameters with large variance (this is an effective evidence of their correct nomination as problematic). Based on the mentioned variance-decomposition analysis some parameters, which are involved in at least one near dependence, could be considered for a preliminary selection as practically unidentifiable candidates.

The computation of the variance-covariance matrix of the regularized biased estimator $\hat{\theta}^{Reg}$ (C^{Reg}) depends on the chosen regularization technique. The general approximation which also gives origin to Eq. (17) is given in Eq. (19) (Bard, 1974; Fessler, 1996). The specific terms used for each regularization technique are given in Table 1.

$$C^{Reg} \approx [H_{\theta}^{Reg}]^{-1} \left[\frac{\partial^2 \Phi^{Reg}}{\partial \theta \partial Y^m} \right] \left[\frac{\partial^2 \Phi^{Reg}}{\partial \theta \partial Y^m} \right]^T [H_{\theta}^{Reg}]^{-1} \quad (19)$$

Using Tikhonov regularization, especially for small regularization parameter values $\lambda \rightarrow 0$ the parameter covariance matrix C^{Tikh} in Eq. (19) can be approximated by $[H_{\theta}^{Tikh}]^{-1}$ (as shown in Table 1). Hereafter we refer to this simplified version as Tikhonov regularization (“Reg” = “Tikh”).

2.4. Optimal experimental design (OED) problem formulation

Improving the parameter precision is the goal of the OED problem given in Eq. (20). Therein, an information function Ψ :

⁸ L-curve analysis is a common tool to determine λ in discrete ill-posed problems (Hansen, 1998). In Section 4.3.3, it is shown that the choice of λ is directly related to the definition of the well-conditioned matrix $\tilde{S}^{Tikh}$ and it thus depends on the ϵ -threshold.

$PSD(Np) \rightarrow \mathbb{R}$ defined on $PSD(Np)$ is minimized (Pukelsheim, 1993). With the domain of Ψ being positive semi-definite the corresponding geometrical shape of those matrices is exploited in optimization. This set is a cone which is convex, pointed, and closed and has interior $PD(Np)$, relative to the space of symmetric matrices $Sym(Np)$ (Pukelsheim, 1993). The control or experimental design vector u is determined so as to optimize Ψ . The vector u minimizes the size of the confidence region of the model parameters. In other words, u maximizes the information content of the experiment.

$$\hat{u}_{crit} = \underset{u}{\operatorname{argmin}} \Psi^{crit}(u, \hat{\theta}) \quad (20)$$

The information function $\Psi^{crit}(u, \hat{\theta})$ is here called OED criterion. The most common OED criteria for full-rank parameter systems (i.e., well-posed PE) are the alphabetic criteria A, D and E such that $\Psi^{crit}, \text{crit} \in \{A, D, E\}$, see Eqs. (21)–(23) (Pukelsheim, 1993). These criteria are appropriate to measure the size of $C(u, \hat{\theta})$ in Eq. (17). Thus, $C(u, \hat{\theta})$ should belong to $PSD(Np)$ as well as H_θ .

$$\Psi^A(u, \hat{\theta}) = \frac{1}{Np} \cdot \operatorname{tr}[C(u, \hat{\theta})] = \frac{1}{Np} \cdot \sum_{i=1}^{Np} \lambda_i(C) = \frac{1}{Np} \cdot \sum_{i=1}^{Np} \frac{1}{(\zeta_i)^2} \quad (21)$$

$$\Psi^D(u, \hat{\theta}) = [\det[C(u, \hat{\theta})]]^{1/Np} = \left[\prod_{i=1}^{Np} \lambda_i(C) \right]^{1/Np} = \left[\prod_{i=1}^{Np} \frac{1}{(\zeta_i)^2} \right]^{1/Np} \quad (22)$$

$$\Psi^E(u, \hat{\theta}) = \lambda_{\max}(C) = \frac{1}{[\zeta_{\min}]^2}, \quad \text{with } \lambda_{\max}(C) = \lambda_{Np}(C), \wedge, \zeta_{\min} = \zeta_{Np} \quad (23)$$

Furthermore, A, D and E-criteria are also called average-variance, generalized variance and minimax criterion, respectively (Pukelsheim, 1993). The most important qualities of each criteria are: the invariance property under re-parameterizations of D (Pukelsheim, 1993), the reduced computational complexity of A (only requiring the computation of the diagonal entries of C) (Pukelsheim, 1993) and the ability to measure ill-posedness of E (Bitterlich and Knabner, 2003).

It can be seen in Eqs. (21)–(23), that according to Eq. (11), the design criteria are related to the eigenvalues of C (i.e., $\lambda_i(C)$, $\forall i = 1, \dots, Np$ in A and D criteria, and, $i = Np$ in E criterion) and therefore to the corresponding singular values of $\tilde{S}$ (i.e., ζ_i , $\forall i = 1, \dots, Np$ in A and D criteria, and, $i = Np$ in E criterion). Note that $\lambda_i(C)$ is the reciprocal of the eigenvalues of H_θ (i.e., $\lambda_i(H_\theta)$, $\forall i = 1, \dots, Np$) (see Eq. (24)). Thus, the squared reciprocal of the singular values of $\tilde{S}$ simplistically approximate the eigenvalues of C of the maximum likelihood estimators (Bitterlich and Knabner, 2003).

$$\lambda_i(C) = \frac{1}{\lambda_i(H_\theta)} = \frac{1}{(\zeta_i)^2} \quad \forall i = 1, \dots, Np \quad (24)$$

Most of the development in OED is carried out for systems of full rank matrices (Pukelsheim, 1993). In this context, the matrix H_θ (and consequently the matrix C) should not be singular. When H_θ is of full rank it belongs to $PD(Np)$. Moreover, the rank of matrix H_θ (as defined in Eq. (8)), is shown to reflect identifiability of the parameter system of interest (Pukelsheim, 1993). Accordingly, a full column rank matrix $\tilde{S}$ (condition for local identifiability) is needed. Otherwise, the calculation of design criteria could be meaningless, for instance, the determinant of H_θ could be zero (with at least one zero-eigenvalue) and then the determinant criterion Φ^D could be indefinite.

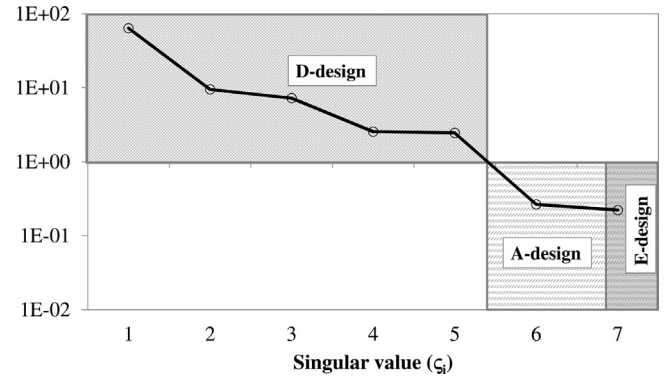


Fig. 3. Influence of different experimental design criteria on the singular value spectrum (SVs) of $\tilde{S}$.

2.5. Graphical interpretation of design criteria and their influence on the singular value spectrum (SVs)

Being H_θ positive definite, the approximation of the confidence region in Eq. (17) has an ellipsoidal shape which is centered at the estimate $\hat{\theta}$ (Bard, 1974). The Np eigenvectors of C are the Np principal axes of the ellipsoid. The longest axis, corresponding to $\lambda_{\max}(C)$ (the largest eigenvalue of C), defines the worst-determined direction in θ space, and the shortest axis (smallest eigenvalue $\lambda_{\min}(C)$) defines the best-determined direction. The target of the design problem in Eq. (20) is to shrink this ellipsoidal region to an acceptable size.

The application of the different design criteria in Eqs. (21)–(23) can be interpreted geometrically (Franceschini and Macchietto, 2008; Schöneberger et al., 2009). The D-criterion aims at minimizing the volume of this confidence region, the A-criterion the dimensions of the enclosing box around the confidence region and the E-criterion the size of its major axis (i.e., $\lambda_{\max}(C)$).

It is worthwhile to point out that each criterion in Eqs. (21)–(23) directly affects the singular values ζ_i for $i = 1, \dots, Np$ of $\tilde{S}$, because of the inverse relation in Eq. (24). The inverse of the square of the smallest singular value of $\tilde{S}$ ($\zeta_{\min} = \zeta_{Np}$) is equal to $\lambda_{\max}(C)$, i.e., the largest eigenvalue of C. Therefore, some or all singular values ζ_i of $\tilde{S}$ are maximized and the SVs of $\tilde{S}$ is, in some way, partially improved when minimizing some eigenvalues of C.

For instance, the A-criterion would preferably act on the smallest singular values (those smaller than 1) because their reciprocal contributes more in the summation of Eq. (21). The multiplicative nature of the D-criterion in Eq. (22) tends to maximize the largest singular values (greater than 1). However, the smallest singular values of $\tilde{S}$ could be indirectly reduced. In this case, $\tilde{S}$ would become worse-conditioned possibly because of an intensification of parameter correlations. This is a well-known feature of the D-optimal criterion (Franceschini and Macchietto, 2008; Schöneberger et al., 2009). Accordingly, the D-design can lead to very large stretched confidence regions with a high correlation of parameters, but to a small volume (Schöneberger et al., 2009). Moreover, the D-design tends to give excessive importance to the parameter to which the model is most sensitive (Franceschini and Macchietto, 2008). (Remember that most of the sensitive parameters are related to the largest singular values of $\tilde{S}$).

Finally, the E-criterion in Eq. (23) increases ζ_{Np} because of its effect on the largest eigenvalue $\lambda_{\max}(C)$. The tendency to improve the small singular values in the SVs of $\tilde{S}$ can therefore drive A- and E- optimal designs to similar solutions. This is especially true for strong correlated parameters, see Schöneberger et al. (2009).

Fig. 3 shows the influence of the different criteria on the SVs of $\tilde{S}$ in presence of singular values smaller than 1. The largest values (top section) of the SVs determine the D-criterion value and the smallest

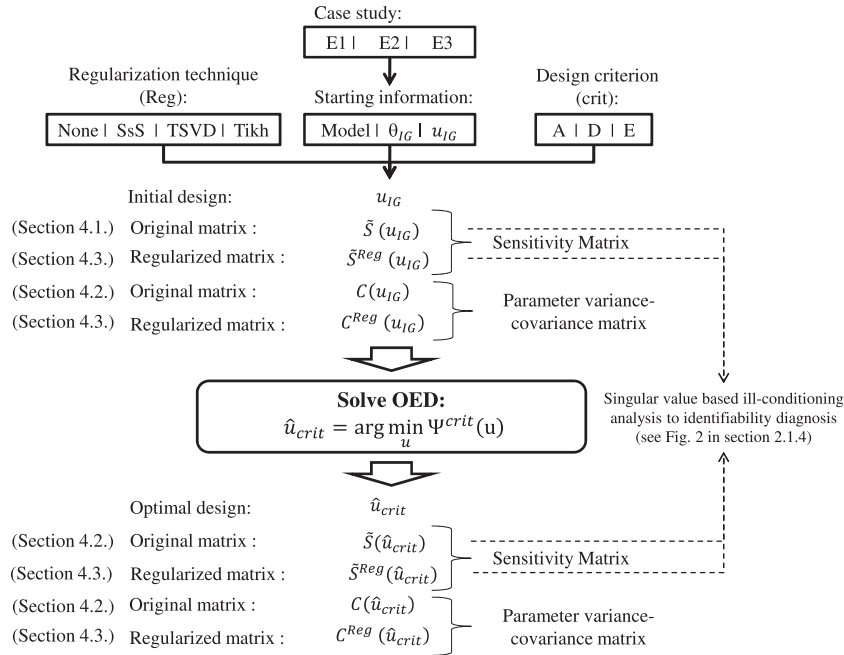


Fig. 4. Main procedure and nomenclature in Section 4.

values (bottom section) the A- and E-criterion values, with the E-criterion being solely determined by the smallest one. Note that the portion of the SVs whose contribution can be neglected is improved only by chance.

It is strongly recommended to carefully analyze the conditioning of $\tilde{S}$ when applying the alphabetic design criteria. For example, the D-criterion cannot be calculated if a SVs of $\tilde{S}$ contains at least one zero-value. That means at least one eigenvalue of H_θ is zero (i.e., $\exists i$ such that $\lambda_i(H_\theta) = 0$), the H_θ is not invertible and hence the covariance matrix C in Eq. (17) cannot be evaluated. Even if the SVs of $\tilde{S}$ does not have exactly zero-singular values, the presence of one or several values near zero could degenerate the inversion and produce criterion values approaching infinity. Moreover, negative eigenvalues could appear because of inherent numerical errors from the computation of H_θ (Eq. (8)) which can affect the positive semi-definiteness of that matrix. Indeed, the theoretically positive semi-definite matrix H_θ (generally all symmetric matrices are positive semi-definite) could lose its positive semi-definite property despite conserving the symmetry in its structure (i.e., $A = A^T$). In those cases, the calculated design criteria are meaningless and an OED should not be performed.

3. Applications

In this work, the effect of ill-posed PE on optimal experimental design is studied for three different case studies E1, E2, E3. These applications are derived from dynamic systems from bioprocessing, being E1: a semi continuous (fed-batch) fermentation reactor (Asprey and Macchietto, 2002); E2: a biochemical growth model in a reactor (Kremling et al., 2004); and E3: a sequencing batch reactor for water treatment (Kaelin et al., 2009; Cruz Bournazou et al., 2012). The degree of complexity of each example increases according to its number of parameters and its state of ill-posedness of the PE from E1 to E3. The starting information (i.e., model, experimental design u_{IG} , parameter initial guess θ_{IG} and true parameters θ^*) is taken from literature when available or defined in this study. For instance, convergence tests on the PE are used to select the

values of θ_{IG} . A brief problem overview and references for detailed information are given in Table 2.

4. Results and discussions

What follows is a numerical analysis of the different nonlinear discrete ill-posed problems E1, E2, E3 referenced in Section 3. The link between identifiability problems and ill-conditioning is established and general problems concerning the solution of OED for ill-posed PE using conventional alphabetic design criteria are discussed. The application of different regularization techniques and the solution of the corresponding modified OED problems are examined.

The remainder of this paper is organized according to the flow diagram depicted in Fig. 4. In Section 4.1 the original state of ill-conditioning and identifiability (see procedure in Section 2.1.4) for each case study E1, E2, E3 at the initial design u_{IG} , i.e., based on $\tilde{S}(u_{IG})$ is established. In Section 4.2 the new state of conditioning and identifiability (see procedure in Section 2.1.4) after the application of OED without regularization at the optimal design $\hat{u}_{crit}$, i.e., based on $\tilde{S}(\hat{u}_{crit})$ is shown. Finally, in Section 4.3 different regularizations methods are applied for the transformation of the original ill-posed problem into a new well-posed (regularized) problem. The regularization is realized for the initial design u_{IG} and the SVs of the regularized sensitivity matrix $\tilde{S}^{Reg}(u_{IG})$ is computed. It is then compared to the SVs obtained for the solution of the regularized OED with the regularized sensitivity matrix $\tilde{S}^{Reg}(\hat{u}_{crit})$ evaluated at the optimal design $\hat{u}_{crit}$. Both sets of SVs are analyzed according to the procedure in Section 2.1.4.

The optimal design $\hat{u}_{crit}$ is obtained from the solution of the OED (see Eq. (20)) for different design criteria with $crit = \{A, D, E\}$ according to Eqs. (21)–(23) and different regularization techniques, with $Reg = \{None, SsS, TSVD, Tikh\}$ (see Table 1).

All numeric computations are performed using MATLAB Release 2013a (The MathWorks Inc., Natick, Massachusetts, United States). Model and parameter sensitivity equations are integrated by CVODES from SUNDIALS (Hindmarsh et al., 2005). PE problems are solved with “lsqnonlin” (nonlinear least squares) function using

Table 2
Problem description for case studies E1, E2 and E3.

Description	Notation	Fed batch fermentation (see Asprey and Macchietto, 2002) Example E1				Biochemical network (see Kremling et al., 2004) Example E2	Activated Sludge Model (ASM3) (see Kaelin et al., 2009) Example E3
		E1	E1a	E1b	E1c		
State variables	$x \in \mathbb{R}^{N_x}$	$[x_1, x_2]$	$[x_1, x_2]$	$[x_1, x_2]$	$[x_1, x_2]$	$[V, B, S, M_1, M_2, M_3]$	$[S_O, S_S, S_{NH}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{ALK}, S_I, X_I, X_H, X_S, X_{STO}, X_{AOB}, X_{NOB}, X_{TSS}]$
Measured state variables	$y \in \mathbb{R}^{N_y}$	$[x_1]$	$[x_1, x_2]$	$[x_1, x_2]$	$[x_1, x_2]$	$[B, S, M_1, M_2, M_3]$	$[S_O, S_S, S_{NH}, S_{NO_2}, S_{NO_3}, S_{ALK}, X_{STO}]$
Data sampling time grid	$t \in \mathbb{R}^{N_m}$	$[2:2:20]$	$[2:2:20]$	$[0.5:0.5:20]$	$[0.25:0.25:20]$	$[2:2:60]$	$[0.002:0.002:0.2]^a$
Initial conditions	$x(t_0) = x_0$	$[5.5, 0.1]$				$[1.0, 0.1008, 1.9134, 0.0620, 0.0079, 0.0749]$	$[0, 1000, 20, 20, 20, 0, 5, 0, 0, 200, 200, 50, 20, 20, 1616.7]$
Input action variables	$u \in \mathbb{R}^{N_u}$	$[u_1, u_2]$				$[q_{in}, q_{out}, c_{in}]$	u_1
Input action time grid	$t \in \mathbb{R}^{N_{mu}}$	$[2 \ 14 \ 20]$				$[20 \ 30 \ 60]$	$[0.02:0.02:0.2]^a$
Input action initial design	$u_{IG} \in \mathbb{R}^{N_u \cdot N_{mu}}$	$u_1 = [0.12, 0.12, 0.12] \ u_2 = [15, 15, 15]$				$u_1 = [0.25 \ 0.35 \ 0.35], u_2 = [0.25 \ 0.35 \ 0.35], u_3 = [2.0 \ 2.0 \ 0.5]$	$u_1 = [0.05, 1.0, 0.05, 0.3, 0.5, 1.0, 0.7, 0.2, 0.05, 1.0]$
Parameters	$\theta \in \mathbb{R}^{N_p}$	$\theta_i, i = 1, \dots, 4$				$\theta_i, i = 1, \dots, 10$ $[r_{1max}, K_S, K_2, KM_1, r_{3max}, KM_2, K_{synmax}, K_{IB}, Y_{XS}, K_{IA}]$ In contrast to Kremling, 2004, where some parameters values are assumed to be exactly known, in this work all parameters are considered as unknown.	$\theta_i, i = 1, \dots, 44$ $[i_{NSS}, i_{NXI}, i_{NXS}, i_{NBM}, f_{XI}, Y_{HO_2}, Y_{HNO_3}, Y_{HNO_2}, Y_{STOO_2}, Y_{STON_3}, Y_{STON_2}, Y_{AOB}, Y_{NOB}, K_H, K_{sto}, \mu_H, \mu_{AOB}, \mu_{NOB}, b_{HO_2}, b_{STOO_2}, b_{AOB}, b_{NOB}, \eta_{HNO_3}, \eta_{HNO_2}, \eta_{HendNO_3}, \eta_{HendNO_2}, \eta_{Nend}, K_X, K_{HO_2}, K_{HO_2inh}, K_{HSS}, K_{HNNH_4}, K_{HNO_3}, K_{HNO_2}, K_{HALK}, K_{HSTO}, K_{AOB_2}, K_{NOB_2}, K_{AOBNH_4}, K_{NOBNO_2}, K_{NALK}, T, K_{Ia20}, S_{OSAT}]$
Parameter initial guess	$\theta_{IG} \in \mathbb{R}^{N_p}$	$[0.5 \ 0.5 \ 0.5 \ 0.5]$				$[7200, 0.133119, 1667700, 3.66, 900000, 3, 0.00246, 0.0498, 0.0000211, 300]^a$	$[0.012, 0.016, 0.012, 0.028, 0.08, 0.32, 0.26, 0.26, 0.32, 0.28, 0.28, 0.072, 0.024, 3.6, 4.8, 1.2, 0.36, 0.26, 0.12, 0.12, 0.06, 0.088, 0.06, 0.06, 0.1, 0.14, 0.04, 0.4, 0.08, 0.08, 4.0, 0.004, 0.2, 0.2, 0.04, 0.04, 0.32, 0.32, 0.056, 0.112, 0.2, 8.0, 400, 41853]^a$
Differential Eq. System		DAE converted to ODE				DAE converted to ODE	

^a Defined by authors.

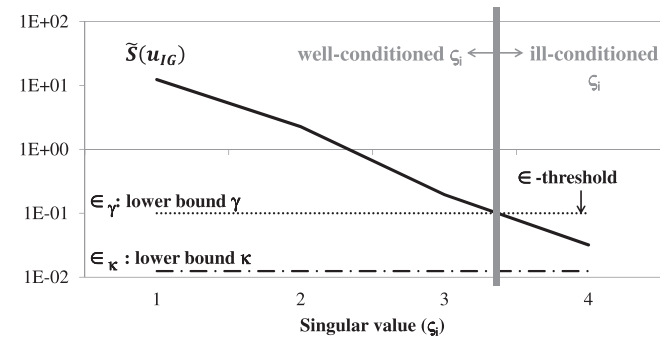


Fig. 5. Singular value spectrum (SVs) of the sensitivity matrix evaluated at the initial design $\tilde{S}(u_{IG})$ for problem E1.

the trust region reflective algorithm. OED problems are solved with “fmincon” (constrained nonlinear multivariable) function using the interior point algorithm. Singular values are computed with “svd” function whereas the eigenvalue system is solved with “eig” function.

4.1. Ill-conditioning analysis for identifiability diagnosis

This section summarizes results from the ill-conditioning analysis for case studies E1, E2 and E3 at the initial design u_{IG} using the sensitivity matrix $\tilde{S}(u_{IG})$. The classification of the ill-conditioning and the selection of singular values generating a well-posed problem are described. The maximum values in Table 3 are used in order to determine the ϵ -threshold to define the numerical ϵ -rank as explained in Section 2.1.3. Remember that $\gamma^{\max}(\tilde{S}) = \gamma^{\max}(S) \cdot \sigma$ and $\kappa^{\max}(\tilde{S}) = \kappa^{\max}(S)$. Here, for the sake of simplicity $\tilde{S}(u_{IG})$, $\kappa(\tilde{S})$, $\gamma(\tilde{S})$, $\kappa^{\max}(\tilde{S})$ and $\gamma^{\max}(\tilde{S})$ are referred to as $\tilde{\kappa}$, $\tilde{\gamma}$, $\tilde{\kappa}^{\max}$ and $\tilde{\gamma}^{\max}$ respectively.

4.1.1. E1 – fed batch fermentation

For this case study, the singular values of the matrix $\tilde{S}$ at u_{IG} vary from $\zeta_4 = 3.1912 \times 10^{-2}$ to $\zeta_1 = 1.2407 \times 10^1$ (see Fig. 5). The condition number and the collinearity index are $\kappa = 3.8879 \times 10^2$ and $\gamma = 3.1376 \times 10^1$, respectively. This problem is categorized as rank-deficient, see features specified in Fig. 1. Note that only one ill-conditioning condition is fulfilled, namely $\gamma > \gamma^{\max}$. The problem is considered slightly ill-posed, $\tilde{S}$ is ill-conditioned because the singular value ζ_4 is relatively close to zero. As can be seen in Fig. 5, ζ_4 passes the ϵ -threshold (defined by $\gamma^{\max}$, i.e., $\epsilon = \epsilon_\gamma$) such that $\zeta_4 < \epsilon$, thus $r_\epsilon = 1$. The model can be considered practically unidentifiable. The variance of all parameters is large (see Appendix 1 Table A.1.1. for details of the parameter variance-decomposition). Furthermore, the ill-conditioned singular value ζ_4 contributes with more than 73% to the variance of all parameters according to Eq. (18). Therefore, the preliminary selection identifies θ as practically unidentifiable. All parameters should be nominated as unidentifiable with inflated variances which should be analyzed carefully. For instance, the variance of the most uncertain parameter $\hat{\theta}_2$ is large compared to the variance of the most precise parameter $\hat{\theta}_4$ ($\text{var}(\hat{\theta}_2) = 74 \text{var}(\hat{\theta}_4)$).

Table 3
Thresholds for condition number $\kappa^{\max}(S)$ and collinearity index $\gamma^{\max}(S)$.

Problem	$\kappa^{\max}(S)$	$\gamma^{\max}(S)$
E1	1000	10
E2	1000	15
E3	1000	15

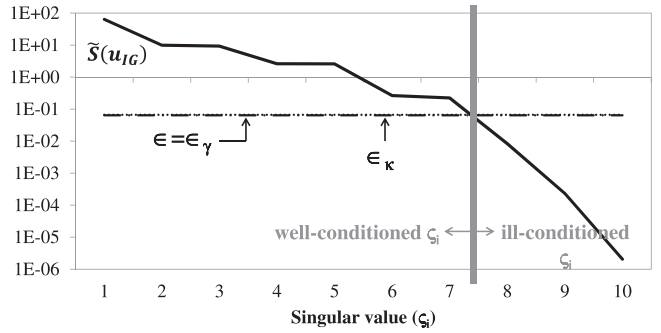


Fig. 6. Singular value spectrum (SVs) of the sensitivity matrix evaluated at the initial design $\tilde{S}(u_{IG})$ for problem E2.

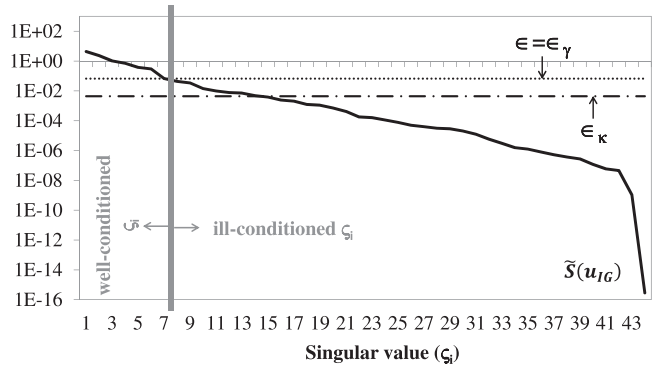


Fig. 7. Singular value spectrum (SVs) of the sensitivity matrix evaluated at the initial design $\tilde{S}(u_{IG})$ for problem E3.

4.1.2. E2 – biochemical network

The singular values of $\tilde{S}$ at u_{IG} vary from $\zeta_{10} = 2.0383 \times 10^{-6}$ to $\zeta_1 = 6.4266 \times 10^1$ with $\kappa = 3.1529 \times 10^7$ and $\gamma = 4.9061 \times 10^5$. With this large condition number and collinearity index the matrix $\tilde{S}$ is categorized as ill-conditioned and thus the problem is ill-posed. This problem can be clearly characterized as rank-deficient. There is a well-determined gap in the SVs of $\tilde{S}$ between large ($\zeta_1 - \zeta_7$) and small singular values ($\zeta_8 - \zeta_{10}$), see Fig. 6. Both ill-conditioning conditions are fulfilled, namely, $\kappa > \kappa^{\max}$ and $\gamma > \gamma^{\max}$. In this case, the condition $\gamma > \gamma^{\max}$ is the stronger condition for the ϵ -threshold (see Eq. (13)), therefore $\epsilon = \epsilon_\gamma$. Only the first seven singular values (i.e., ζ_i for $1 \leq i \leq 7$) are larger than ϵ , thus they are considered well-conditioned and $r_\epsilon = 7$. The model is therefore considered practically unidentifiable. According to Eq. (18), the most precise parameter $\hat{\theta}_9$ has a variance of 0.15 (only parameter with variance less than 1) whilst the most imprecise parameter $\hat{\theta}_{10}$ has a variance of 2.4066×10^{11} (see Appendix 1 Table A.1.2. for details of the parameter variance-decomposition). Parameters $\hat{\theta}_5$ and $\hat{\theta}_6$ also have enormous variances with values larger than 2.6×10^7 . Moreover, the ill-conditioned singular values $\zeta_8, \dots, \zeta_{10}$ contribute with more than 57% to the variance of $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \hat{\theta}_5, \hat{\theta}_6, \hat{\theta}_7, \hat{\theta}_8$ and $\hat{\theta}_{10}$. Thus, these parameters are preliminary selected as practically unidentifiable. A further analysis with SsS will show the refinement of this selection (see Section 4.3.1).

4.1.3. E3 – ASM3

The singular values of $\tilde{S}$ at u_{IG} vary from $\zeta_{44} = 2.8427 \times 10^{-16}$ to $\zeta_1 = 4.3461$ (see Fig. 7).⁹ Compared to E1 and E2, problem E3 shows the largest condition number and collinearity index,

⁹ It is important to point out that ζ_{44} is reaching the machine precision which indicates that this value is not reliable, but demonstrates the gravity of the ill-posedness of the problem.

with $\kappa = 1.5289 \times 10^{16}$ and $\gamma = 3.5178 \times 10^{15}$. The singular values gradually decay to zero which indicates an ill-determined rank problem. Hence, it has the biggest severity of ill-posedness. Only the first seven singular values (i.e., ς_i for $1 \leq i \leq 7$) correspond to a well-conditioned matrix ($r_\epsilon = 7$). The selection of these well-conditioned singular values is based on $\epsilon = \epsilon_\gamma$. As expected, this model is categorized as practically unidentifiable. Any parameter has a variance less than (corresponding to the most precise parameter $\hat{\theta}_{14}$, i.e., $\text{var}(\hat{\theta}_{14}) = 1.23 \times 10^6$, see Appendix 1 Table A.1.3. for details of the parameter variance-decomposition). All parameter variances are highly influenced¹⁰ by the ill-conditioned singular values and therefore, none of them are reliable. Moreover, the parameter variances are inflated masking the individual identifiability of parameters and making a preliminary parameter selection impossible. Thus, a more elaborate analysis is applied to determine the identifiable parameters according to SsS in Section 2.2.1 (see results in Section 4.3.1). More effects of this severe ill-posedness on computations of OED criteria without regularization are discussed in Section 4.2.

Similarly as in E1 (Section 4.1.1), in E3 the presence of small singular values (defined by $\epsilon = \epsilon_\gamma$) strongly reduces the number of well-conditioned singular values and is an indication for nearly linearly interdependencies rather than for numerical instabilities (see Section 2.1.2). Moreover, if those small singular values are considered well-conditioned, the consequences are overestimation of parameter variances, erroneous uncertainty quantification and improper results of the identifiability analysis.

4.2. Optimal design without regularization

In this section the results for the optimal design without regularization are discussed. For all problems E1–E3, the application of the A-, D- and E- criteria yield the expected characteristic changes in the SVs (according to Section 2.5). Generally, A and E-designs focus on the lower part of the SVs whereas the target of the D-design is mostly directed to raise the top section of the SVs. The values in Table 3 are used to define the ϵ -threshold.

4.2.1. E1 – fed batch fermentation

For this case, A and E-designs reach almost the same optimal experimental solution thus, almost the same SVs (see Fig. 8a). In Fig. 8a, those designs generate a lift of the bottom section. This indicates a variance reduction for all parameters because of the increment of the smallest singular values (e.g. $\varsigma_4(\tilde{S}(\hat{u}_A)) = 1.6735 \times 10^{-1}$) and $\varsigma_4(\tilde{S}(\hat{u}_E)) = 1.6775 \times 10^{-1}$). For instance, the variance of the most uncertain parameter $\hat{\theta}_2$ is substantially reduced with respect to the most precise parameter $\hat{\theta}_4$ (i.e. $\text{var}(\hat{\theta}_2) = 8.6\text{var}(\hat{\theta}_4)$). A and E-designs also yield SVs with the smallest collinearity index ($\gamma(\tilde{S}(\hat{u}_A)) = 5.9756$ and $\gamma(\tilde{S}(\hat{u}_E)) = 5.9611$) and the ill-conditioned singular values are completely removed from the problem according to the ϵ -threshold (see in Fig. 8a that neither the spectrum of $\tilde{S}(\hat{u}_A)$ nor $\tilde{S}(\hat{u}_E)$ are intercepted by the bound ϵ). On the contrary, for the D-design the ill-conditioning remains with only three singular values being well-conditioned (see Fig. 8b). Nevertheless, the lifting of the SVs observed in Fig. 8b (especially in the bottom section) promotes parameter variance reduction. All parameters have at least 50% of variance reduction (e.g., the most precise parameter $\hat{\theta}_1$ has 88% variance reduction). Unfortunately, the most uncertain

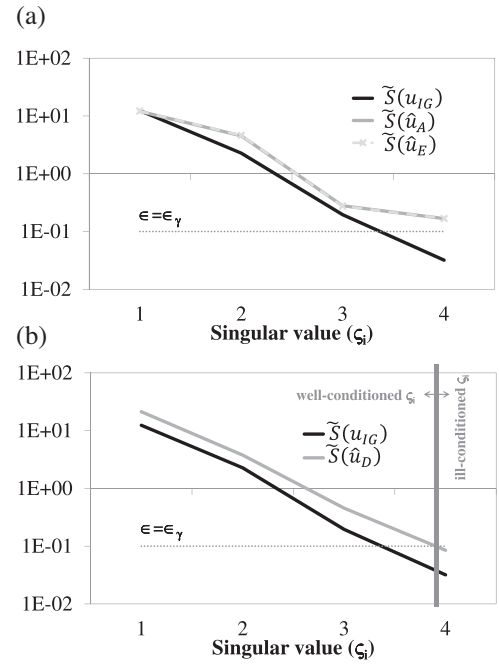


Fig. 8. Change in the singular value spectrum (SVs) of the sensitivity matrix $\tilde{S}$ for the OED without regularization. Results are shown for the initial design $\tilde{S}(u_{IG})$ and optimal experiment design for problem E1 with: (a) A-design $\tilde{S}(\hat{u}_A)$ and E-design $\tilde{S}(\hat{u}_E)$, and (b) D-design $\tilde{S}(\hat{u}_D)$.

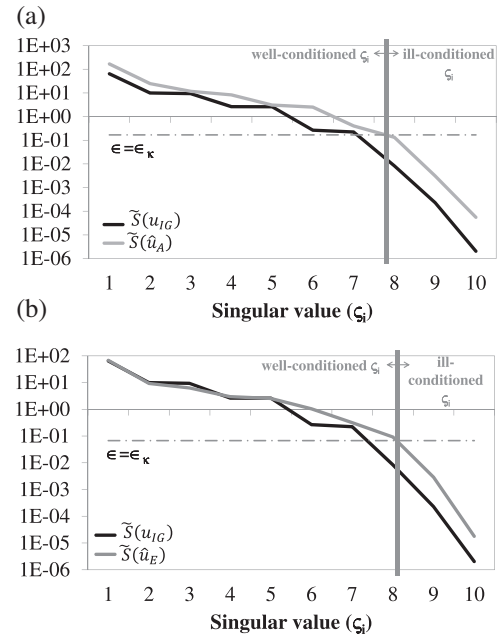


Fig. 9. Change in the singular value spectrum (SVs) of the sensitivity matrix $\tilde{S}$ for the OED without regularization. Results are shown for the initial design $\tilde{S}(u_{IG})$ and optimal experiment design for problem E2 with: (a) A-design $\tilde{S}(\hat{u}_A)$ and (b) E-design $\tilde{S}(\hat{u}_E)$. Note that the lower bound ϵ_κ is computed for the SVs of $\tilde{S}(\hat{u}_{crit})$.

parameter $\hat{\theta}_2$ remains with large variance ($\text{var}(\hat{\theta}_2) = 43\text{var}(\hat{\theta}_1)$) and it is still considered unidentifiable.

4.2.2. E2 – biochemical network

For this case, Fig. 9 shows the change in the SVs for the A- and E-optimal design. In Fig. 9a, it can be seen that the A-design generates a lift of the whole spectrum, although the main elevation can be observed in the bottom section. This indicates a variance reduction

¹⁰ In Appendix 1 Table A.1.3 it is possible to see that 21 parameters have individual variance-decomposition proportions greater than 0.5 ($\pi_i \geq 0.5$) for the most ill-conditioned singular values ($\pi_{40}, \dots, \pi_{44}$). However, the complete set of ill-conditioned singular values ($\pi_8, \dots, \pi_{44}$) contribute almost totally with the variance of the 44 parameters.

for all parameters. For the E-design in Fig. 9b only the bottom section of the spectrum is lifted. Indeed, the A-design yields the best results in terms of condition number and collinearity index with $\kappa(\tilde{S}(\hat{u}_A)) = 3.0198 \times 10^7$ and $\gamma(\tilde{S}(\hat{u}_A)) = 1.7995 \times 10^4$ which correspond to $\varsigma_{10} = 5.5570 \times 10^{-5}$ and $\varsigma_1 = 1.6781 \times 10^2$. However, those values still indicate ill-conditioning of $\tilde{S}(\hat{u}_A)$ as $\kappa > \kappa^{\max}(\tilde{S})$ and $\gamma > \gamma^{\max}(\tilde{S})$, and thus the presence of unidentifiable parameters. The number of well-conditioned singular values remains seven ($\varsigma_1 - \varsigma_7$), the same as for the initial design. For the A-design, $\kappa^{\max}(\tilde{S})$ is the stronger condition for the ϵ -threshold (see Eq. (13)), i.e., $\epsilon = \epsilon_\kappa$. In Fig. 9a, the A-criterion (average variance, see Eq. (21)) is reduced from $\Psi^A(u_{IG}) = 2.4071 \times 10^{10}$ to $\Psi^A(\hat{u}_A) = 3.2393 \times 10^7$. For the A-design, the most precise parameter $\hat{\theta}_1$ has a variance of 2.6323×10^{-3} whilst the most imprecise parameter $\hat{\theta}_{10}$ has 3.2383×10^8 . Moreover, parameters $\hat{\theta}_5$ and $\hat{\theta}_6$ also have enormous variances larger than 5×10^4 . For $\hat{\theta}_3, \hat{\theta}_4, \hat{\theta}_5, \hat{\theta}_6, \hat{\theta}_7, \hat{\theta}_8$, and $\hat{\theta}_{10}$ the ill-conditioned singular values ($\varsigma_8 - \varsigma_{10}$) contribute with more than 65% to their variance.

For the E-optimal design the results are not better. This is mainly because of the big (numerical) difficulty in minimizing the large value of the design criterion generated by the nearness to zero of the smallest singular values ($\Psi^E(u_{IG}) = 1/\varsigma_{10}^2 = 2.4069 \times 10^{11}$). Accordingly the E-criterion could not be further reduced, with $\Psi^E(\hat{u}_E) = 3.1409 \times 10^9$ and the SVs of $\tilde{S}(\hat{u}_E)$ varies from $\varsigma_{10} = 1.7843 \times 10^{-5}$ to $\varsigma_1 = 6.7548 \times 10^1$ still indicating an ill-conditioned matrix $\tilde{S}(\hat{u}_E)$. With this design eight well-conditioned singular values are found. After the optimal design, the most precise parameter $\hat{\theta}_1$ has a variance of 1.0304×10^{-1} whilst the most imprecise parameter $\hat{\theta}_{10}$ has 3.1407×10^9 according to Eq. (18). Moreover, parameters $\hat{\theta}_5$ and $\hat{\theta}_6$ also have enormous variances larger than 1×10^5 . For $\hat{\theta}_5, \hat{\theta}_6$ and $\hat{\theta}_{10}$ the ill-conditioned singular values ($\varsigma_9 - \varsigma_{10}$) contribute with more than 99.9% of their variance. In general, there is an improvement in parameter variance because of the increment of the small singular values. However, also the E-design does not produce a well-posed problem, i.e., a problem free of identifiability problems.

4.2.3. E3 – ASM3

The severe ill-posedness of problem E3 (see Section 4.1.3) makes a reliable computation of an OED impossible. The computation of the eigensystem of $H_\theta(u_{IG})$ and $C(u_{IG})$ does not give reasonable results. For $H_\theta(u_{IG})$ one negative eigenvalue is obtained whereas eight negative eigenvalues are found for $C(u_{IG})$. The reasons are numerical errors (related to the machine precision), coming from matrix operations (matrix product in Eq. (8) and inversion in Eq. (17)) and the solution of the ill-conditioned matrix eigensystem. It is obvious that the computed negative eigenvalues are illogical for the positive semi-definite matrices H_θ and C .

Still, while a D-criterion value cannot be computed because of an error in the applied algorithm, A- and E-criterion values can be obtained. To illustrate the magnitude of ill-conditioned OED criteria, the singular values ς_i for $i = 1, \dots, Np$ of $\tilde{S}$ are used to compute A and E-criteria according to Eqs. (21) and (23), respectively. Those values read $\Psi^A(u_{IG}) = 2.8 \times 10^{29}$ and $\Psi^E(u_{IG}) = 1.2 \times 10^{31}$, respectively. It has to be noted, that the same criteria have values of $\Psi^A(u_{IG}) = 1.8 \times 10^{16}$ and $\Psi^E(u_{IG}) = 8.8 \times 10^{17}$ when the eigenvalues λ_i for $i = 1, \dots, Np$ of $C(u_{IG})$ are taken into consideration. In general, the criteria computed from the singular values of $\tilde{S}$ are by definition more reliable. They also capture better the high singularity of the matrix C (e.g., presence of singular values near the machine precision, i.e., 10^{-16}). It is important to note that under these circumstances a gradient-based minimization is impossible. Eventually computed results are not trustworthy and it is not advisable to draw conclusions about the identifiability of parameters for this kind of severe ill-conditioned problems.

To summarize, for all studied problems the ill-conditioning of $\tilde{S}$ cannot be overcome by solving the optimal design problem without regularization. Although the ill-conditioning is sometimes improved (reduction of the parameter variance), in none of the problems this is sufficient for guaranteeing stable PE and practically identifiability.

4.3. Optimal design with regularization

In this section the solution of the regularized OED problem is discussed. The common feature for all regularizations techniques is the generation of a new better-conditioned sensitivity matrix for the initial design $\tilde{S}^{\text{Reg}}(u_{IG})$, $\forall \text{Reg} \in \{\text{SsS}, \text{TSVD}, \text{Tikh}\}$, i.e., the ill-conditioning is partially or completely removed depending on the selected value of the regularization parameter, see Section 4.3.3. This is illustrated for the difference between the original SVs of $\tilde{S}^{\text{None}}(u_{IG})$ and new regularized SVs of $\tilde{S}^{\text{Reg}}(u_{IG})$, where ill-conditioned singular values have been removed either by parameter space reduction (Reg = SsS) or truncation (Reg = TSVD) or have been corrected to higher values (Reg = Tikh).

For the determination of the well-conditioned singular values of $\tilde{S}$ (via the numerical ϵ -rank, see Section 2.2.1) the thresholds in Table 3 are used.

4.3.1. Parameter subset selection (Reg = SsS)

For the SsS (see Section 2.2.1) the following features are observed:

1. For all studied examples E1–3, at the initial design u_{IG} , only a subset of the parameter space (determined by the rank of $\tilde{S}(u_{IG})$) is identifiable. Accordingly, only three parameters in example E1 (θ_4, θ_1 and θ_3) and seven parameters in example E2 ($\theta_1, \theta_3, \theta_5, \theta_9, \theta_7, \theta_4, \theta_2$) and seven parameters in example E3 ($\theta_{23}, \theta_{44}, \theta_{15}, \theta_{12}, \theta_{16}, \theta_{14}, \theta_8$) are considered identifiable. The new reduced problems (i.e., regularized problems) promote a well-conditioned sensitivity matrix $\tilde{S}^{\text{SsS}}(u_{IG})$, that means all singular values in its SVs are well-conditioned. Thus, even for the strongly ill-conditioned problem E3, the OED could be successfully computed for all considered design criteria.

For problem E2 and E3, the SVs of the regularized matrices $\tilde{S}^{\text{SsS}}(u_{IG})$ and $\tilde{S}^{\text{SsS}}(\hat{u}_{\text{crit}})$ are shown in Fig. 10. For totally linear independent parameters each singular value of $\tilde{S}$ would match exactly one parameter. As this is not the case, each singular value is related to several parameters. Accordingly, the SVs of the regularized matrix $\tilde{S}^{\text{SsS}}(u_{IG})$ is not identical to the SVs of the original matrix $\tilde{S}(u_{IG})$. Moreover, it can be seen, in Fig. 10, that those singular values which belong to the reduced matrix $\tilde{S}^{\text{SsS}}$ take equal or smaller values than those coming from the original matrix $\tilde{S}$ (interlacing inequality for singular values). Nevertheless, the tendency (for rank-deficient problems) to conserve the largest singular values associated with the well-conditioned parameters is evident (see Fig. 10a for Problem E2).

2. The regularized OED improves only the active parameter subset and thus, only the associated singular values. In Fig. 10 it can be seen how the reduced SVs is lifted (shifted to higher values) from $\tilde{S}^{\text{SsS}}(u_{IG})$ to $\tilde{S}^{\text{SsS}}(\hat{u}_{\text{crit}})$.

Despite of the improvements in the conditioning of the reduced matrix $\tilde{S}^{\text{SsS}}(\hat{u}_{\text{crit}})$, for the optimal design the conditioning of the original matrix $\tilde{S}(\hat{u}_{\text{crit}})$ is not always improved, e.g., for problem E1, all optimal designs generates SVs with higher condition number and collinearity index. For the D-design the following values are obtained: $\kappa(\tilde{S}(\hat{u}_D)) = 8.4 \times 10^2 > \kappa(\tilde{S}(u_{IG}))$ and $\gamma(\tilde{S}(\hat{u}_D)) = 3.6 \times 10^1 > \gamma(\tilde{S}(u_{IG}))$.

For example E2, the original matrix $\tilde{S}(\hat{u}_A)$ for the A-optimal design continues being ill-conditioned with $\gamma(\tilde{S}(\hat{u}_A)) = 1.9 \times 10^4 > \gamma^{\max}(\tilde{S})$. An increment in ς_1 and ς_{10} of $\tilde{S}(\hat{u}_A)$ produces a

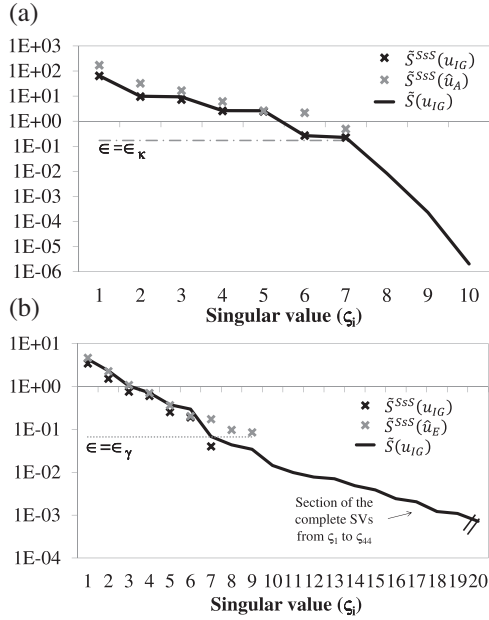


Fig. 10. Singular value spectrum (SVs) of the original and the regularized sensitivity matrix after applying SsS, $\tilde{S}$ and $\tilde{S}^{SsS}$ respectively. Results are shown for the initial and optimal experimental design, u_{IG} and $\hat{u}_{crit}$, respectively, for problem: (a) E2 where $\hat{u}_{crit} = \hat{u}_A$ and (b) E3 where $\hat{u}_{crit} = \hat{u}_E$. The solid-black curve shows the SVs of the original sensitivity matrix without regularization at the initial design $\tilde{S}(u_{IG})$. The black-cross and gray-cross markers show the SVs of the regularized (reduced) matrix at the initial design $\tilde{S}^{SsS}(u_{IG})$ and at the optimal design $\tilde{S}^{SsS}(\hat{u}_{crit})$, respectively. Note that the lower bound ϵ_κ is computed for the SVs of $\tilde{S}^{SsS}(\hat{u}_{crit})$.

reduced condition number $\kappa(\tilde{S}(\hat{u}_A)) = 3.2 \times 10^6 < \kappa(\tilde{S}(u_{IG}))$. But still $\kappa(\tilde{S}(\hat{u}_A)) > \kappa^{\max}(\tilde{S})$ and the rank of the problem remains seven. Accordingly, the ill-conditioning is improved but the problem still remains ill-posed.

3. In some cases the A and E-optimal designs could improve the problem rank and attract new parameters to the identifiable region.

This is shown for problem E3 with the E-design in Fig. 10b, where the problem rank improves from 7 to 9. The improvement of the SVs of $\tilde{S}^{SsS}$ (from u_{IG} to $\hat{u}_E$) moves ζ_8 and ζ_9 into the feasible region defined by $\epsilon = \epsilon_\gamma$. The condition number and collinearity index are slightly reduced $\kappa(\tilde{S}(\hat{u}_E)) = 1.4365 \times 10^{16} < \kappa(\tilde{S}(u_{IG})) = 1.5289 \times 10^{16}$ and $\gamma(\tilde{S}(\hat{u}_E)) = 2.3338 \times 10^{15} < \gamma(\tilde{S}(u_{IG})) = 3.5178 \times 10^{15}$, respectively. However, the matrix $\tilde{S}(\hat{u}_E)$ conserves its severe ill-conditioning. The same increment in the active parameter subset dimension is observed for the A-optimal design.

4.3.2. Truncated singular value decomposition (Reg = TSVD)

For the TSVD (see Section 2.2.2) the following features are observed:

1. In contrast to the indirect manipulation of the SVs in SsS, in TSVD the SVs is directly manipulated (truncated) and the remaining singular values do not change. The truncated spectrum is then used to construct (approximate) a new matrix $\tilde{S}^{TSVD}$.
2. The TSVD produces a well-conditioned but rank-deficient matrix $\tilde{S}^{TSVD}(u_{IG})$. The variance-covariance matrix $C^{TSVD}(u_{IG})$ (see Table 1) is then also rank-deficient and thus, the D-optimal criterion cannot be computed. Hence, only A and E-designs are studied for TSVD.
3. In Fig. 11, the SVs for the A and E-designs for problem E2 and E3 are depicted. From the comparison of Figs. 10 and 11 it can be seen, that for both, TSVD and SsS based OED, very similar results

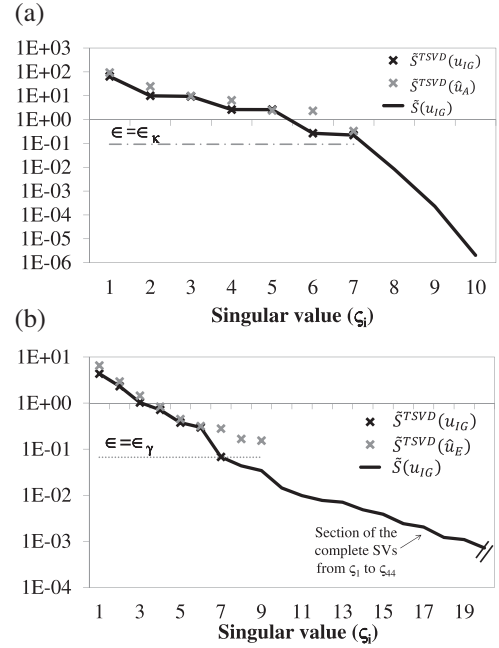


Fig. 11. Singular value spectrum (SVs) of the original and the regularized sensitivity matrix after applying TSVD, $\tilde{S}$ and $\tilde{S}^{TSVD}$, respectively. Results are shown at the initial and optimal experimental design, u_{IG} and $\hat{u}_{crit}$, respectively, for problem: (a) E2 where $\hat{u}_{crit} = \hat{u}_A$ and (b) E3 where $\hat{u}_{crit} = \hat{u}_E$. The solid-black curve shows the SVs of the original sensitivity matrix without regularization at the initial design $\tilde{S}(u_{IG})$. The black-cross and gray-cross markers show the SVs of the regularized (approximated) matrix at the initial design $\tilde{S}^{TSVD}(u_{IG})$ and at the optimized design $\tilde{S}^{TSVD}(\hat{u}_{crit})$, respectively. Note that the lower bound ϵ_κ is computed for the SVs of $\tilde{S}^{TSVD}(\hat{u}_{crit})$.

are obtained. This is true for the problem rank (number of identifiable parameters/singular values) as well as for the influence of the different design criteria (see also Fig. 3). However, both, the computed design criteria and the optimal design vector differ in their values according to the regularization technique applied. This is certainly due to the differences in the number of considered parameters (whole parameter space for TSVD, reduced space for SsS) and the differences in the values of the initial SVs (compare Figs. 10 and 11). It has also to be noted that the design criteria in Eqs. (21)–(23) are scaled differently, with N_p and r_ϵ for TSVD and SsS, respectively.

4.3.3. Tikhonov regularization (Reg = Tikh)

For the Tikhonov regularization (see Section 2.2.3) two different regularization parameters (see Eq. (16)) are tested, i.e., $\lambda_1 = 0.001$ (weak regularization) and $\lambda_2 = 0.1$ (strong regularization). The following features are observed:

1. The Tikhonov regularization yields an increment (lifting) of the SVs of the matrix $\tilde{S}(u_{IG})$ at the initial design maintaining the original parameter space dimension and the number of singular values. For a stronger regularization parameter (i.e., $\lambda = \lambda_2$) the transformation of the SVs is also stronger (compare Fig. 12a and Fig. 12b). The lifting of the SVs yields a transformation of the bottom section of the SVs by fixation of the small singular values to λ^2 . Accordingly, singular values less than the square value of the respective regularization parameter λ^2 are approximately replaced by λ^2 , i.e., $\forall i \text{ } \zeta_i < \lambda^2 \text{ then } \zeta_i \approx \lambda^2$.

For Problem E2 with $\lambda = \lambda_1$, the effect of the regularization is very small and the SVs of $\tilde{S}^{Tikh-\lambda_1}$ almost remains identical to that shown in Fig. 6. This can be expected, taking into account that the smallest singular value of $\tilde{S}$, $\zeta_{10} = 2.0383 \times 10^{-6}$ is already larger than $\lambda_1^2 = 1.0 \times 10^{-6}$. Nevertheless, a reduction of 10% in the condition number and collinearity index is

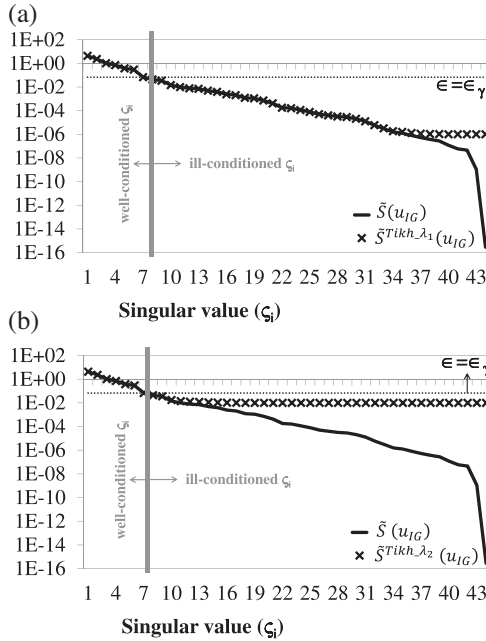


Fig. 12. Singular value spectrum (SVs) of the original $\tilde{S}$ and regularized sensitivity matrix $\tilde{S}^{Tikh}$ for problem E3 with: (a) $\lambda_1 = 0.001$ (weak regularization) and (b) $\lambda_2 = 0.1$ (strong regularization). The solid-black curve shows the SVs of the original sensitivity matrix without regularization at the initial design $\tilde{S}(u_{IG})$. The black-cross markers show the SVs of the regularized matrix at the initial design $\tilde{S}^{Tikh}(u_{IG})$.

achieved. On the other hand, choosing a stronger regularization with $\lambda = \lambda_2$, the last three singular values of $\tilde{S}$ are increased to values around $\lambda_2^2 = 1.0 \times 10^{-2}$ in $\tilde{S}^{Tikh, \lambda_2}(u_{IG})$ (see Fig. 12b). Therefore, $\kappa(\tilde{S}^{Tikh, \lambda_2}(u_{IG}))$ and $\gamma(\tilde{S}^{Tikh, \lambda_2}(u_{IG}))$ are drastically decreased to 6.4×10^3 and 1.0×10^2 , respectively. Despite of this improvement, the number of well-conditioned singular values remains seven (because $\varsigma_i \approx \lambda_2^2 < \epsilon = 6.67 \times 10^{-2}$ for $i = 8, \dots, 10$) demonstrating that the problem still remains ill-posed.

For Problem E3 with $\lambda = \lambda_1$, the smallest nine singular values are increased (see Fig. 12a), i.e., $\varsigma_i \approx \lambda_1^2$ for $i = 36, \dots, 44$. However, the regularization is not strong enough. Hence the matrix C^{Tikh, λ_1} still has three negative eigenvalues. For the stronger regularization with $\lambda = \lambda_2$, the smallest thirty five singular values are increased (see Fig. 12b), i.e., $\varsigma_i \approx \lambda_2^2$ for $i = 10, \dots, 44$. The new regularized matrix C^{Tikh, λ_2} has positive eigenvalues and the new condition number and collinearity index are $\kappa(\tilde{S}^{Tikh, \lambda_2}(u_{IG})) = 4.3461 \times 10^2$ and $\gamma(\tilde{S}^{Tikh, \lambda_2}(u_{IG})) = 1.0 \times 10^2$. Again the ill-conditioning improvement is not enough to turn the problem in a well-conditioned one (the same seven well-conditioned singular values) however its transformation is significant. It must be noted that values of λ larger than ϵ , which completely remove the ill-conditioning, could be selected. However, those values strongly affect the problem fitting producing large residuals. Thus they are not considered in this study.

- For problem E1 and E2, A and D optimal designs are successfully computed, see e.g. Fig. 13. Moreover, the influences of the different criteria are as described in Section 2.5, see e.g., the results for E2 for the A-design in Fig. 13a.

For Problem E3 with $\lambda = \lambda_1$, the D-design cannot be computed as its criterion is infinite at u_{IG} . In contrast, for the stronger regularization with $\lambda = \lambda_2$ the D-design can be computed and the criterion is reduced from $\Psi^D(u_{IG}) = 2.2 \times 10^3$ to $\Psi^D(\hat{u}_D) = 1.6 \times 10^3$.

- Generally, the computed E- designs are not always reliable especially for strong regularization with $\lambda = \lambda_2$. Here the smallest

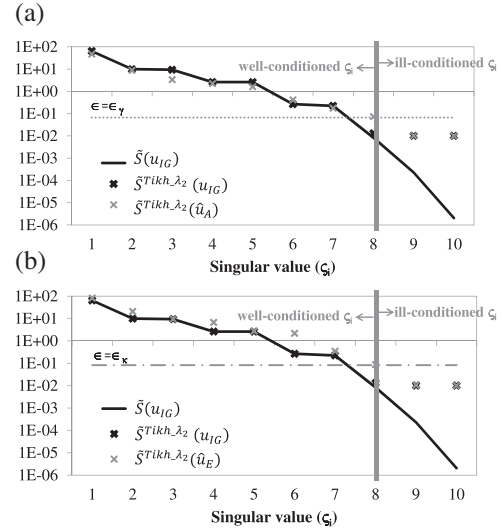


Fig. 13. Singular value spectrum (SVs) of the original and the regularized sensitivity matrix after applying Tikh, $\tilde{S}$ and $\tilde{S}^{Tikh}$, respectively. Results are shown at the initial and optimal experimental design, u_{IG} and $\hat{u}_{crit}$, respectively, for problem E2: (a) $\hat{u}_{crit} = \hat{u}_A$ and (b) $\hat{u}_{crit} = \hat{u}_E$. The solid-black curve shows the SVs of the original sensitivity matrix without regularization at the initial design $\tilde{S}(u_{IG})$. The black-cross and gray-cross markers show the SVs of the regularized matrix at the initial design $\tilde{S}^{Tikh}(u_{IG})$ and at the optimal design, respectively. Note that the lower bound ϵ_κ is computed for the SVs of $\tilde{S}^{Tikh}(\hat{u}_{crit})$.

singular values of $\tilde{S}^{Tikh}(\hat{u}_{crit})$ are repeatedly corrected in each optimization iteration (i.e., $\varsigma_i \approx \lambda^2 \forall i \varsigma_i < \lambda^2$). Accordingly, if the optimal design does not promote a SVs with a smallest singular value ς_{Np} larger than λ^2 , this singular value is always $\varsigma_{Np} \approx \lambda^2$. This situation generates an almost constant E-criterion value, equal to $1/(\lambda^2)^2$, and makes the optimization impossible.

5. Summary and conclusions

In model-based parameter estimation and experimental design the formulation of well-posed problems is crucial for the numerical computation of stable and unique solutions. Thus, it is strongly advisable to perform the relatively simple local analysis of the conditioning of the sensitivity matrix (S). Moreover, if an ill-posed problem is identified, its problem type, either rank deficient or of ill-determined rank, and problem severity should be assessed. Important indicators suggested in this contribution are the singular value spectrum, condition number, and collinearity index of S .

There exists a relationship between those singular values in the SVs of S and commonly used metrics for parameter identifiability and OED, namely the eigenvalues of the H_θ (FIM) and the eigenvalues of the parameter covariance-variance matrix (C) of the parameter estimate. This information is particularly useful when dealing with ill-posed experimental design problems, as it is highly recommendable to do the numerical analysis and implementation based on the singular values of S . Additionally, a graphical interpretation of the influence of the alphabetic experimental design criteria applied to the SVs has been presented. It turns out that A- and E-optimal criteria mainly improve the smallest singular values of S while the D-optimal criterion improves the largest singular values. Thus, the potential of an experimental design for improving the parameter precision can be similarly analyzed as the well-known graphical interpretation of the influence of the alphabetic design criteria applied to the confidence region of the parameter estimates represented by C .

However, for ill-posed problems with ill-conditioned S and consequently H_θ and C , the direct application of the alphabetic design criteria can lead to numerical instabilities, especially in

the eigensystem of H_θ . Whereas the computation of the SVs is numerically stable also for ill-posed problems, small singular values of S (especially those near zero) have a large influence on the design criterion evaluated for H_θ and C . They can produce huge criterion values, which then complicate or even impede an appropriate optimization. Thus, a control of the smallest singular values of S is needed (e.g., maximizing the smallest singular values) and by this, a control of the most uncertain parameters (those related to the largest eigenvalues of C).

As a solution to this, different regularization techniques, namely SsS, TSVD and Tikhonov, were presented together with details on their implementation. To illustrate their application to parameter identification and OED problems, three different case studies of increasing complexity and ill-posedness were considered. The following particular features, strengths and weaknesses of each technique were identified.

- Each regularization technique implies a transformation of the original problem. Hence, the regularized optimal design certainly improves the ill-conditioning of the regularized problem but not necessarily the ill-conditioning of the original one. Moreover, the methods ensure to obtain a solution but provide no information whether the obtained solution is still useful in the original context.
- Particular attention should be paid to ill-determined rank problems. Because of the nearness to zero of the singular values, a strong regularization is needed. Unfortunately, this pushes the regularized problem away from the original problem and increases the bias in the solution.
- The effect of each regularization technique modifies the problem to improve its ill-conditioning which is evidenced in the SVs of the matrix S of the regularized problem. In SsS, the problem is transformed excluding the unidentifiable parameters and by this the smallest (ill-conditioned) singular values. In TSVD the ill-conditioned singular values are substituted by zero and the new problem only considers singular values with large magnitude in the original problem. Finally, the effect of Tikhonov regularization is to fix the ill-conditioned singular values to the magnitude of the squared regularization parameter (λ^2), whilst the large singular values are not largely modified.
- The selection of the regularization parameter, i.e., ϵ -threshold for SsS and TSVD, and the scalar parameter λ in Tikhonov, is problem dependent and should be done carefully. Firstly, the ill-conditioning of the problem should be improved and, secondly, the problem should not be completely changed. For instance, in Tikhonov, the regularization parameter λ should improve the ill-conditioning in the sensitivity matrix (which should also control the solution norm of the PE) without greatly affecting the essence of the original problem (controlling the norm of the residual in the PE). In that sense, the value of λ^2 should be as large as the first singular value which is considered to cause the ill-conditioning.
- When applying Tikhonov regularization, a computed E-optimal design is not viable, if this new design does not promote a SVs with the smallest singular value larger than λ^2 . Here, the regularization parameter continues fixing the small singular values (i.e., $\zeta_{Np} \approx \lambda^2$) and makes an optimization impossible (fixing the E-criterion value to $1/(\lambda^2)^2$ in each iteration).
- For SsS and TSVD there exists the tendency to conserve the largest singular values which are associated to the well-conditioned parameters. From an applications point of view, the SsS seems the most natural approach, as the regularization acts in the original parameter space. It preserves the physical meaning of parameters and provides useful information on the number of identifiable parameters as well as on the ranking of parameters regarding their linear independence and sensitivity.

- However, it has to be noted, that a change in the dimension of the identifiable parameter subset (in SsS) or the number of well-conditioned singular values (in TSVD) introduces a discontinuity in the evaluation of the design criterion. This problem is similar to the inherent non-differentiability in the definition of the E-optimal criterion, where a possible switching in the smallest eigenvalue introduces this behavior.
- Generally it might be worthwhile to note, that any increment in the SVs (promoted by the OED) does not automatically mean an improvement in the ill-posedness of the problem. On the contrary, large values in the top section of the SVs (without an adequate increase in the rest) may lead to higher condition number which then give rise to ill-estimated parameters for unstable PE. This behavior was observed here especially for the D-design.

Conflict of interest

The authors have declared no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.compchemeng.2015.03.002>

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